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INTERNATIONAL STANDARD



**Explosive atmospheres –
Part 30-2: Electrical resistance trace heating – Application guide for design,
installation and maintenance**



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**Explosive atmospheres –
Part 30-2: Electrical resistance trace heating – Application guide for design,
installation and maintenance**

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

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International Standard IEC/IEEE 60079-30-2 has been prepared by IEC technical committee 31: Equipment for explosive atmospheres, in cooperation with the Petroleum & Chemical Industry Committee of the IEEE Industrial Applications Society under the IEC/IEEE Dual Logo Agreement.

NOTE A list of IEEE participants can be found at the following URL:
http://standards.ieee.org/downloads/60079/60079-30-2-2015/60079-30-2-2015_wg-participants.pdf.

This first edition of IEC/IEEE 60079-30-2 cancels and replaces the first edition of IEC 60079-30-2 published in 2007 and constitutes a technical revision.

This edition includes the following significant changes, apart from a general review and updating of the first edition of IEC 60079-30-2, harmonization with IEEE Std 515, with respect to the previous edition:

- the relocation of trace heater product design methodology and requirements to IEC/IEEE 60079-30-1;
- the relocation and/or duplication of information on installation, maintenance, and repair to the MTs under SC31J for their addition into IEC 60079-14, IEC 60079-17, and IEC 60079-19;
- the inclusion of more detailed information on safety showers and eyewash units;
- the introduction of Annexes from IEEE Std 515.

The significance of changes between IEC 60079-30-2, Edition 1.0 (2007) and IEC/IEEE 60079-30-2, Edition 1.0 (2014) is as listed below:

		Type		
Changes	Clause	Minor and editorial changes	Extension	Major technical changes
Addition of clarification for the exclusion of areas coverage classifications of EPLs Ga and Da	1	X		
Addition of requirements for the Division method of area classification that may be applied by some users	1			C1
Relocation of heat loss design requirements to IEC/IEEE 60079-30-1	6.3	X		
Addition of safety shower and eyewash station design requirements	6.16			C2
Addition of Annex for an example of a design data record	Annex A	X		
Addition of Annex for a checklist of installation requirements	Annex B	X		
Addition of Annex for an example of a trace heater commissioning record	Annex C	X		
Addition of Annex for an example of a maintenance schedule and log record	Annex D	X		
Addition of Annex for pipe heat loss considerations	Annex E	X		
Addition of Annex for vessel heat loss considerations	Annex F	X		
Addition of Annex for heat up and cool down considerations	Annex G	X		
Addition of Annex for a method to determine the equivalent thickness of insulating cements	Annex H	X		

NOTE The technical changes referred to include the significance of technical changes in the revised IEC Standard, but they do not form an exhaustive list of all modifications from the previous version.

Explanations:

A) Definitions

Minor and editorial changes

clarification
decrease of technical requirements
minor technical change
editorial corrections

These are changes which modify requirements in an editorial or a minor technical way. They include changes of the wording to clarify technical requirements without any technical change, or a reduction in level of existing requirement.

Extension

addition of technical options

These are changes which add new or modify existing technical requirements, in a way that new options are given, but without increasing requirements for equipment that was fully compliant with the previous standard. Therefore, these will not have to be considered for products in conformity with the preceding edition.

Major technical changes

addition of technical requirements
increase of technical requirements

These are changes to technical requirements (addition, increase of the level or removal) made in a way that a product in conformity with the preceding edition will not always be able to fulfil the requirements given in the later edition. These changes have to be considered for products in conformity with the preceding edition. For these changes additional information is provided in clause B) below.

NOTE These changes represent current technological knowledge. However, these changes should not normally have an influence on equipment already placed on the market.

B) Information about the background of 'Major Technical Changes'

C1 – The requirements for the Division method of area classification are applicable only for users of this standard intending qualification for these areas.

C2 – The design requirements for safety showers and eyewash units have been included for harmonization and for added safety.

The text of this standard is based on the following IEC documents:

FDIS	Report on voting
31/1190/FDIS	31/1199/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

International standards are drafted in accordance with the ISO/IEC Directives, Part 2.

This standard is intended to be used in conjunction with IEC/IEEE 60079-30-1:2014, *Explosive atmospheres – Part 30-1: Electrical resistance trace heating – General and testing requirements*.

A list of all parts of IEC 60079 series, under the general title *Explosive atmospheres*, can be found on the IEC website.

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- withdrawn,
- replaced by a revised edition, or
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EXPLOSIVE ATMOSPHERES –

Part 30-2: Electrical resistance trace heating – Application guide for design, installation and maintenance

1 Scope

This part of IEC 60079 provides guidance for the application of electrical resistance trace heating systems in areas where explosive atmospheres may be present, with the exclusion of those classified as requiring EPL Ga/Da (traditional relationship to Zone 0 and Zone 20 respectively). This standard also provides guidance for explosive atmospheres incorporating the Division method of area classification that may be applied by some users of this standard.

NOTE Information on the Division method is given in NFPA 70 and CSA C22.1.

It provides recommendations for the design, installation, maintenance and repair of trace heating systems including associated control and monitoring equipment. It does not cover devices that operate by induction heating, skin effect heating or direct pipeline heating, nor those intended for stress relieving.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-426, *International Electrotechnical Vocabulary – Part 426: Equipment for explosive atmospheres*

IEC 60079-0, *Explosive atmospheres – Part 0: Equipment – General requirements*

IEC 60079-15, *Explosive atmospheres – Part 15: Equipment protection by type of protection “n”*

IEC/IEEE 60079-30-1, *Explosive atmospheres – Part 30-1: Electrical resistance trace heating – General and testing requirements*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60050-426, IEC 60079-0 and IEC/IEEE 60079-30-1 apply.

4 Application considerations

4.1 General

This part of IEC 60079 supplements the requirements specified in IEC 60079-14, IEC 60079-17 and IEC/IEEE 60079-30-1.

Where trace heating systems are to be installed in explosive atmospheres, full details of the area classifications shall be specified. The specification shall state, as applicable, the

required equipment protection levels Gb, Gc, Db, and Dc (traditional relationship to Zone 1, Zone 2, Zone 21, and Zone 22 respectively), and/or the Division 1 and Division 2 explosive atmospheres, the gas or dust groups, and temperature classification. Where special considerations apply or where site conditions may be especially onerous, these conditions shall be detailed in the trace heating specification.

The specification for heating systems to be installed on mobile equipment or skid units (e.g. re-locatable structures) should accommodate the adverse conditions in which the trace heating system may be used.

Where any parts of the trace heating system are likely to be exposed, those parts should be suitable for the environment.

4.2 Corrosive areas

All components of electric trace heating systems should be examined to verify that they are compatible with any corrosive materials that may be encountered during the lifetime of the system. Trace heating systems operating in corrosive environments have a higher potential for failure than in non-corrosive environments. Deterioration of the thermal insulation system is made worse by corrosion of the weather barrier and the possibility of moisture leaks soaking the thermal insulation.

4.3 Process temperature accuracy

4.3.1 Type I

A Type I process is one for which the temperature should be maintained above a minimum point. Ambient sensing control may be acceptable. Large blocks of power may be controlled by means of a single control device and an electrical distribution panel board. Heat input may be provided unnecessarily at times and wide temperature excursions should be tolerable. Energy efficiency may be improved through the use of dead-leg control or ambient proportional control techniques (see 6.14).

4.3.2 Type II

A Type II process is one for which the temperature should be maintained within a moderate band. Control by mechanical thermostats is typical.

4.3.3 Type III

A Type III process is one for which the temperature should be controlled within a narrow band. Electronic controllers using thermocouple or resistance-temperature detector (RTD) process temperature sensors facilitate field (work site) calibration and provide maximum flexibility in the selection of temperature alarm and monitoring functions. Heat input capability may be provided to preheat an empty pipe or raise the fluid temperature, or both, within a specified range and time interval. Type III systems require strict adherence to flow patterns and thermal insulation systems.

4.4 Installation considerations

If failure of any part of the trace heating system can result in a safety or process problem, then the trace heating system may be considered to be a critical component of the total process. The temperature control and circuit monitoring requirements of an application may be defined according to the temperature control types described in 4.3.

When trace heating is critical to the process, circuit monitoring for correct operation, malfunction alarms, and back-up trace heaters should be considered. Spare or back-up controllers can be specified to be automatically activated in the event of a fault being indicated by the monitoring/alarm system. Back-up trace heaters may allow maintenance or repairs to be performed without a process shutdown and may be used to enhance reliability.

5 Thermal insulation

5.1 General

The selection, installation and maintenance of thermal insulation is a key component in the performance of an electrical trace heating system. The thermal insulation system is normally designed to limit heat loss with the trace heating system compensating for the remainder. Therefore, problems with thermal insulation have a direct impact on the overall system performance.

The primary function of thermal insulation is to reduce the rate of heat transfer from a surface that is operating at a temperature other than ambient. This reduction of energy loss may:

- reduce operating expenses;
- improve system performance;
- increase system output capability.

Prior to any heat loss analysis for an electrically traced pipeline, vessel or other mechanical equipment, a review of the selection of the insulation system is recommended. The principal areas for consideration are as follows:

- selection of an insulation material;
- selection of a weather barrier (cladding);
- selection of the economic insulation thickness with consideration for optimum trace heater design;
- selection of the proper insulation size.

Information about the equivalent thickness of insulating cements is given in Annex H.

5.2 Selection of insulating material

The following are important aspects to be considered when selecting an insulation material. These factors should be considered and the selection optimised according to the operator's criteria:

- temperature rating;
- thermal conductivity, λ , of the insulation;
- mechanical properties;
- chemical compatibility and corrosion resistance;
- moisture resistance;
- health risks during installation;
- fire resistance;
- toxicological properties when exposed to fire;
- costs.

Insulation materials commonly available include:

- expanded silica;
- mineral fibre;
- cellular glass;
- urethane;
- fibreglass;
- calcium silicate;

- polyisocyanurate;
- perlite silicate.

For soft insulants (mineral fibre, fibreglass, etc.), actual pipe size insulation may be used in many cases by banding the insulation tightly. Care should be taken to prevent the trace heater from being buried within the insulation, which may cause damage to the trace heater or may restrict proper heat transfer. As an alternative, the next largest pipe size insulation that can easily enclose pipe and electric trace heater is also acceptable. Rigid insulation (calcium silicate, expanded silica, cellular glass, etc.), may be pipe-size insulation if board sections are cut to fit the longitudinal joint. This type of installation technique is commonly referred to as an extended leg installation. Alternatively, the next largest insulation size may be selected to accommodate the trace heater. In all cases, the insulation size and thickness should be clearly specified.

Insulation for valves, flanges, pumps, instruments, and other irregularly shaped equipment may be constructed for the particular configuration. This can be fabricated from block, insulation segments or flexible removable covers.

Non-insulated or partially insulated pipe supports or equipment require additional heat input to compensate for the higher heat loss. Insulating cements or fibrous materials should be used to fill cracks and joints. Where insulating cements are used for total insulation of an irregular surface, a proportionally thicker layer of insulating cement may be applied to achieve the desired insulating capability.

5.3 Selection of weather barrier (cladding)

Proper operation of an electrically trace heated system depends upon the insulation being dry. Electric tracing normally has insufficient heat output to dry wet thermal insulation. Some insulation materials, even though removed from the piping and force dried, never regain their initial characteristics after once being wet.

Straight piping may be weather-protected with metal jacketing, polymeric, or a mastic system. When metal jacketing is used, it should be smooth with formed, modified “S” longitudinal joints. The circumferential end joints should be sealed with closure bands and supplied with sealant on the outer edge or where they overlap (see Figure 1).

Jacketing that is overlapped or otherwise closed without sealant is not effective as a barrier to moisture. A single, unsealed joint can allow a considerable amount of water to leak into the insulation during a rainstorm, therefore specifications for elastomeric water sealing of cladding should be suitable for long term exposure to the environmental conditions.

The type of weather barrier used should, as a minimum, be based on a consideration of the following:

- effectiveness in excluding moisture;
- corrosive nature of chemicals in the area;
- fire protection requirements;
- durability to mechanical abuse;
- cost.

5.4 Selection of economical thickness to provide optimum trace heating design

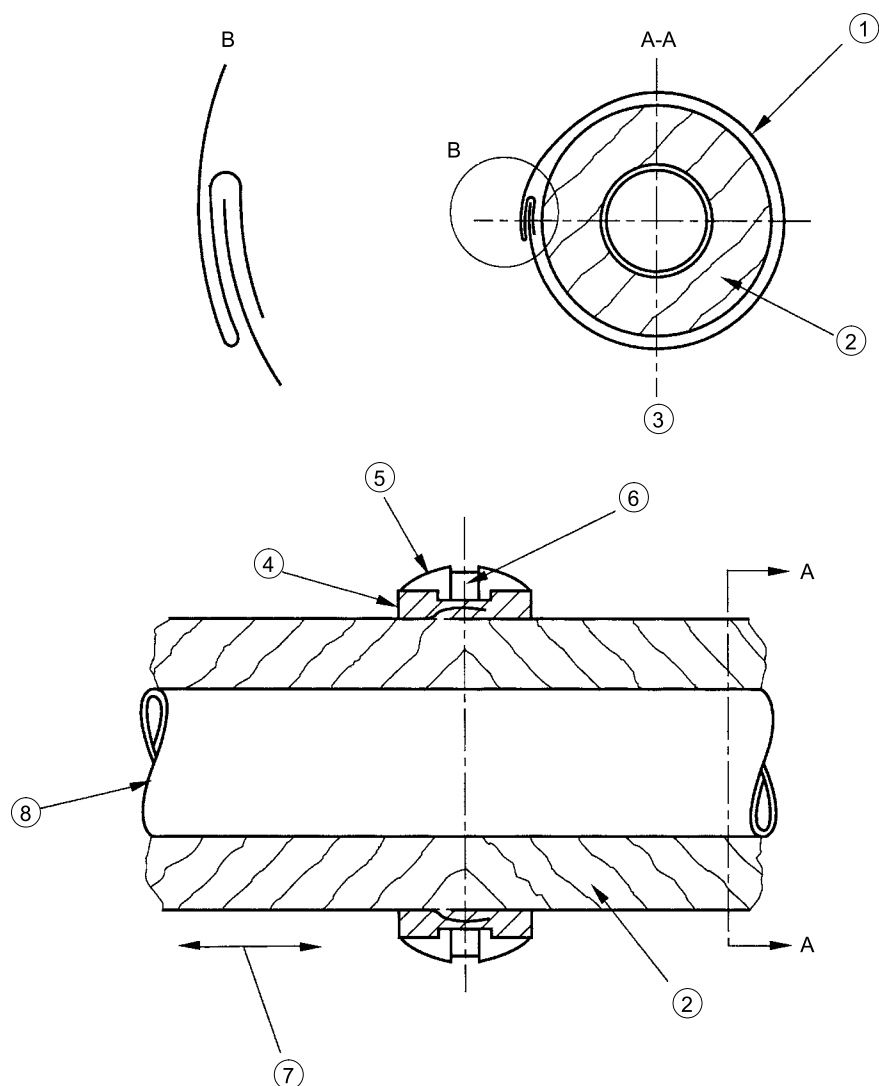
At a minimum, an economic consideration of the insulation weighs the initial costs of the materials and installation against the energy saved over the life of the insulation. It should be noted that the actual insulation thicknesses do not always correspond exactly to the nominal insulation thickness. When choosing the insulation size, considerations should be made as to whether or not the actual pipe-size insulation is suitable for accommodating both pipe and trace heater.

The insulation thickness selected for economic consideration of non-traced portions of a system does not necessarily provide for an optimum or acceptable trace heating system design and installation. Consideration of the electric trace heating design should be made in conjunction with the selection of insulation. Selection of an optimum insulation thickness for electric traced lines can do the following:

- reduce the power output level necessary for each trace heater;
- increase the length of trace heater circuits;
- decrease the number of circuits required;
- decrease the total power required for electric trace heater systems.

5.5 Double insulation

The double insulation technique may be employed when the pipe temperature exceeds the maximum allowable temperature of the trace heater. Prevention of the freezing of condensate in high-temperature steam lines when these lines are not in use is a typical application. It consists of locating the trace heater between two layers of insulation surrounding the pipe. The essence of the double-insulation technique is to determine the correct combination of inner and outer insulation type and thickness that results in an acceptable interface temperature for the trace heater. This relationship is illustrated in Figure 2. Note that maximum ambient temperature conditions should be considered in this determination. Consideration should be given to joint installation and alignment of the inner insulation to account for thermal expansion of the piping.

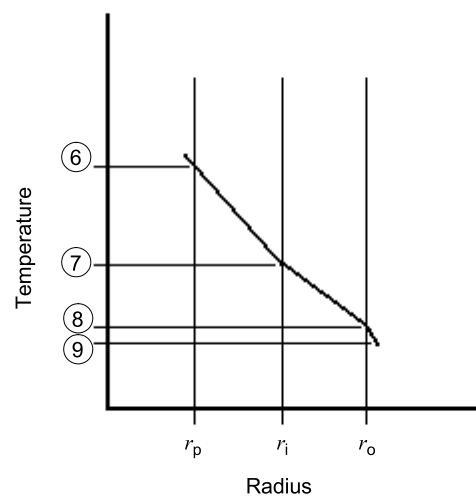
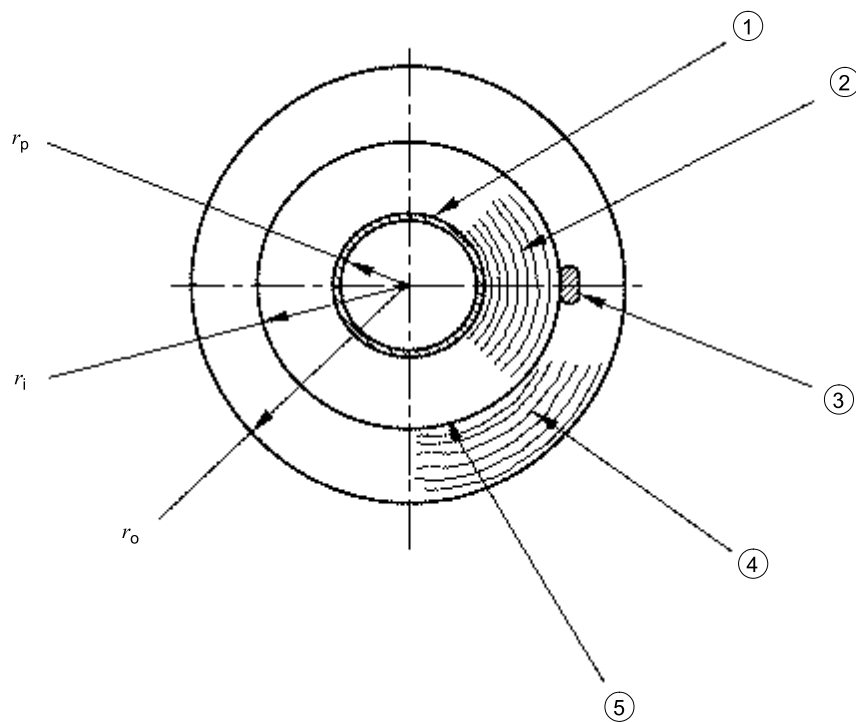


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Key

- | | |
|-------------------------------|-------------------|
| 1 metal jacket | 5 closure band |
| 2 insulation | 6 insulated strap |
| 3 metal jacket insulated pipe | 7 movement |
| 4 mastic sealer | 8 pipe |

Figure 1 – Thermal insulation – Weather-barrier installation



IEC

Key

- | | |
|--------------------------|--|
| 1 pipe | 6 maximum temperature pipe |
| 2 inner insulation layer | 7 interface temperature |
| 3 trace heater | 8 outer insulation surface temperature |
| 4 outer insulation layer | 9 ambient temperature |
| 5 metal foil (aluminium) | |

Figure 2 – Typical temperature profile

6 System design

6.1 General

Each trace heating application imposes unique demands on the designer to achieve the desired temperature and maintain it within the specified conditions. Trace heating systems necessarily interface with other specified items of equipment such as thermal insulation and the electrical supply available to power the system. The final system is an integration of all these component parts so the values of these interface items have to be known and controlled in order to design systems that perform as required.

The design of any trace heating system should conform to all IEC/IEEE requirements for the use of electrical equipment and with the requirements of this standard. Consideration should be given as to the maintenance of the systems and process equipment, to energy efficiency, and to testing the installed systems for operational satisfaction and safety.

Considerations that affect the efficiency of trace heating design are insulation thickness and supply voltage. Insulation type selection is generally made based on temperature and environmental conditions. The insulation thickness selected for traced lines should be made in conjunction with the trace heater selection, to provide an optimum trace heating design. See 5.4 for additional discussion of this topic. Selection of the supply voltage to be applied to trace heating circuits can greatly influence the overall power supply and economics of the system.

It is necessary to maintain adequate distances between the pipes and between pipes and parts of the structure in order to permit installation of the thermal insulation.

When designing trace heating systems for use in explosive atmospheres, additional constraints are imposed due to the requirements and classification of the area under consideration.

The details of each circuit design should be recorded on a form such as the example indicated in Table A.1 and retained.

Information about design for heat loss calculations and examples is provided in Annex E for pipes and in Annex F for vessels.

6.2 Purpose of, and major requirement for, trace heating

Trace heaters should be selected and installed so as to provide sufficient power for:

- a) compensation of heat loss when maintaining a specified temperature of a workpiece at the specified minimum ambient temperature; see calculation method in IEC/IEEE 60079-30-1; or
- b) raising the temperature of a workpiece, and its contents when specified, within a specified time period; see calculation method in 6.6; or
- c) a combination of a) and b).

The trace heater selection shall consider the determination of the maximum possible sheath temperature under adverse conditions as specified in IEC/IEEE 60079-30-1. The methods employed include the product classification method, stabilized design, and controlled design.

The temperature may be reduced, for example, through adjustments to the system parameters, by the use of multiple tracers to reduce the power produced per unit length, or by the selection of the temperature control system. The excess of installed power over the power as required, and the way in which trace heaters are applied, installed and operated shall not be the cause, not even after evaluation of worst-case conditions, of any unacceptable risk in explosive atmospheres.

The manufacturer of the trace heating is required to establish the design calculations and methods. The end user is constrained to the design calculations and methods specified by the manufacturer.

6.3 Training

Persons involved in the design of electric trace heating systems should be suitably trained in all techniques required.

6.4 Selection of trace heater

6.4.1 General

For a particular application, there are some basic design characteristics that should be reviewed to determine the choice of trace heater(s). These are as follows.

- a) The maximum withstand temperature of the trace heaters shall be greater than the maximum possible workpiece temperature (which may be greater than the normal maintain temperature).
- b) Trace heaters should be suitable for operation in the environmental conditions specified, for example, a corrosive atmosphere or a minimum ambient temperature.
- c) Trace heaters shall be certified for use in the particular explosive atmosphere.

For any application, there is a maximum allowable power density at which a trace heater can be used without damaging either the workpiece or its contents. This is particularly critical in certain cases, such as with lined pipes, vessels containing caustic soda or with heat sensitive materials. The maximum allowable power density shall be recorded in the system documentation. Multiple tracing or spiralling of a single trace heater may be required.

6.4.2 Site-fabricated trace heaters

Site-fabricated trace heaters are permissible provided:

- installation personnel are trained in the special techniques required;
- trace heaters are certified for site-fabrication;
- the installation instructions supplied by the manufacturer cover site-fabrication;
- trace heater(s) pass the field (site work) tests specified in 8.5.5;
- trace heater(s) is/are marked in accordance with IEC/IEEE 60079-30-1.

Trace heater(s) not excluded by the above considerations are technically suitable for the application, but it is still necessary to determine the maximum allowable power density of each. This is a function of the construction, the maximum withstand temperature, the maximum operating temperature and maximum permissible temperature of the workpiece and thermal insulation.

For each particular trace heater, the maximum allowable power density shall be determined from the manufacturer's data that is based on tests specified in IEC/IEEE 60079-30-1. The value used shall be chosen so that the maximum withstand temperature is not exceeded. The limiting value of maximum allowable power density for each trace heater is either the value chosen from the manufacturer's data or that specified for the process, whichever is the lower. However, the power density may be further limited by the need for multiple tracing.

The designer may then select the type, length or size and loading of the trace heater. The actual installed load should be not less than the design loading and the actual power density not greater than that obtained above. The type of trace heater and the values of installed load and power density should be recorded in the system documentation.

6.4.3 Specific types of trace heating

The series and parallel types of trace heating are generally defined by their electrical characteristics.

Series trace heaters typically use the electrical conductor as the heating element, so the voltage supply and circuit length become critical parameters in the design of each circuit. Series trace heaters with polymeric insulation are particularly suited for installations with long circuit lengths. Series trace heaters with mineral insulation (MI) and metallic sheaths are particularly suited for very high process temperature maintenance. Parallel trace heaters typically consist of two parallel conductors with a separate polymeric or metallic heating element that draws voltage from the conductors. These are characteristically used for freeze protection, and for process temperature maintenance of complex piping installations. The constant wattage type typically has a spirally wound metallic heating element.

The PTC type typically consists of a polymeric heating element extruded between the conductors. The power-limiting type typically falls between the previous types, with higher output at higher operating temperatures than the PTC type, and with lower operating temperatures than the constant wattage type.

6.5 Maximum temperature determination

6.5.1 General

It is important that the maximum sheath temperatures of trace heaters are determined accurately for all applications in explosive atmospheres. It is imperative that the circuit designer follows all guidelines, specifications, and design methodology provided by the manufacturer in accordance with IEC/IEEE 60079-30-1. Some examples of why it is important to determine the maximum sheath temperature of the trace heater are:

- a) applications such as non-metallic piping where the maximum temperature of the trace heater may approach the maximum withstand temperature of the workpiece, the thermal insulation, or other components of the system;
- b) applications that have no control or ambient sensing control, and that have potentially high tracer sheath temperatures at equilibrium;
- c) critical applications where a high degree of process temperature accuracy is required;
- d) applications in explosive atmospheres where it is assumed that the controlling thermostat is not functioning, such that the tracer sheath temperature cannot exceed the ignition temperature or T-rating of the gas, vapour, or combustible dust expected in that location.

It is important that the circuit designer understands that there are three possible measures for protecting the tracing system from exceeding the high-limit temperature – the PTC characteristic of the trace heater, the use of a limiter or control device, and by stabilized design calculations. The methods and limitations for the evaluation of maximum temperature are provided in IEC/IEEE 60079-30-1.

When multiple trace heaters (especially on installations with multiple flow conditions) are grouped together under a single temperature control device, each design condition shall be analysed as a stabilized design.

6.5.2 PTC characteristic

Trace heaters that significantly reduce power with an increase in temperature, i.e. positive temperature coefficient trace heaters, can be assigned temperature classes by testing (as given in IEC/IEEE 60079-30-1). In many applications further temperature limiting control measures are not necessary, provided that the temperature class of the trace heater is lower in temperature than that specified for the application. Nevertheless the limiter and stabilized design measures can be applied to operate the system in a narrower band of process temperatures.

6.5.3 Stabilized design

Stabilized design is based on the principle of determining the maximum workpiece and trace heater sheath temperatures under a worst-case set of conditions. This is a calculation of the equilibrium conditions that occur when the heat input equals the system heat loss. The worst-case set of conditions include:

- a) maximum ambient temperature, typically assumed to be 40 °C unless otherwise specified;
- b) no wind (still air);
- c) use of a conservative or minimum value for the thermal conductivity of the thermal insulation;
- d) no temperature control by design or to simulate a failed temperature controller;
- e) the trace heater is operated at its stated operating voltage plus 10 %;
- f) the trace heater is assumed to be operating at the upper limit of the manufacturing tolerance, or at the minimum specific resistance for series trace heaters.

Specific requirements for stabilized designs are found in IEC/IEEE 60079-30-1. Testing for stabilized design is also defined in IEC/IEEE 60079-30-1. Typically the maximum sheath temperature of the trace heater is calculated from formulae derived from the evaluation of empirical data and by the theoretical approach described in the Annex of IEC/IEEE 60079-30-1 for the trace heater product design verification methodology. Alternatively, design programs that calculate the maximum sheath temperature on the basis of these worst-case parameters may be used.

6.5.4 Controlled design

A temperature limiter/control device shall prevent the trace heater from exceeding the high-limit temperature by typically sensing:

- a) the temperature of the workpiece or other application components;
- b) the temperature on the sheath of the trace heater;
- c) parameters other than temperature, such as current.

The temperature limiter/control system shall open the circuit if a malfunction occurs in the sensor or the limiter/control device. Specific requirements for controlled designs are found in IEC/IEEE 60079-30-1.

A control and monitoring system shall be used which meets the application requirements of this standard that is appropriate to the application considerations. Refer to Clause 7.

6.6 Heat up and cool down considerations

In certain plant operations, it may be necessary to specify that the trace heating system is capable of raising the temperature of a static product within a certain time period, or that it may be necessary to consider cool down times. Heat up and cool down considerations are given in Annex G.

6.7 Design information

6.7.1 Design information documentation

To help ensure a workable trace heating design, the design function should be furnished with up-to-date process and piping information and should be updated with any specification and drawing revisions that pertain to the trace heating system. Any or all of the following may be applicable:

- a) Process parameters:
 - 1) Area classification documentation including applicable ignition temperatures

- 2) System flow diagram
- 3) Equipment detail drawings (pumps, valves, strainers, etc.)
- 4) Process procedures that would cause elevated pipe temperatures, that is, steam out or exothermic reactions
- b) Piping parameters:
 - 1) Equipment layout drawings (plans, sections, etc.)
 - 2) Pipe drawings (plans, isometrics, line lists, etc.)
 - 3) Piping specifications
 - 4) It is necessary to maintain adequate distances between the pipes and between pipes and parts of the structure in order to permit installation of the thermal insulation.
- c) Thermal insulation requirements and data:
 - 1) Thermal design parameters
 - 2) Thermal insulation specifications
 - 3) Thermal insulation schedules
 - 4) Bills of material
- d) Electrical parameters:
 - 1) Electrical drawings (one lines, elementaries, etc.)
 - 2) Electrical equipment specifications
 - 3) Equipment details
 - 4) Equipment installation and instruction manuals
 - 5) Bills of material

6.7.2 Isometric or trace heater configuration line lists and load charts

Each trace heater circuit design should include the following information:

- a) Design data:
 - 1) Area classification, including the lowest ignition temperature for each area if applicable
 - 2) Piping designation or line number
 - 3) Pipe size, rating and material
 - 4) Thermal insulation type, nominal size, thickness, and *k*-factor
 - 5) Trace heater designation and/or circuit number
 - 6) Temperature to be maintained
 - 7) Maximum process temperature
 - 8) Minimum ambient temperature
 - 9) Maximum exposure temperature (when applicable)
 - 10) Maximum sheath temperature (when required)
 - 11) Heat loss at desired maintenance temperature per unit length of pipe
 - 12) Heat up parameters (when required)
 - 13) Watts per unit length of trace heater at desired maintenance temperature
 - 14) Watts, total
 - 15) Circuit current, start-up and steady state
 - 16) Bills of material
- b) Installation drawings:
 - 1) Isometric equipment installation drawings

- 2) Piping location with line numbers
- 3) Location of power connections, end seals, and temperature sensors as applicable
- 4) Length of piping
- 5) Trace ratio of trace heater per length of pipe
- 6) Length of trace heaters
- 7) Extra length of trace heater applied to valves, pipe supports, and other heat sinks
- 8) Operating voltage
- 9) Power distribution panel number or designation, the alarm and control equipment designation, and set points

6.8 Power system

The power system for an electric trace heating system consists of the power source, distribution transformers, and the heat tracer distribution system. Refer to IEC 60364-1 for specific information.

Since the power source is a key factor in the overall design of an electric trace heating system, it is recommended that voltage levels and physical location be determined in the early stages of the design.

The kilovoltampere (kVA) ratings of the distribution transformers should be based on the maximum design operating load, plus consideration for circuit startup and selected spare capacity. Maximum design operating load and startup loads should be based on minimum startup temperature. To optimize transformer size, consider startup of the fewest possible circuits at a time. Transformer kVA rating should be at least 125 % of computed maximum load due to trace heater currents during cold start up. A power system may require several distribution transformers, depending on the magnitude and physical distribution of the trace heating loads. The distribution transformer secondary should be an earthed system.

The physical location of the distribution system should be considered in relation to the piping system when developing the distribution system. Whenever practical, locate the panels in unclassified and noncorrosive areas. The selection of the distribution panel enclosure is based on the environment and area classifications. All enclosures should have the specific system identification clearly marked on the outside, and the circuit directories should be easily accessible. The branch circuit wiring of each trace heating circuit requires an over-current protective device. Selection of branch circuit protective device ratings should be based on heater start-up currents and their duration at the minimum temperature the heating device may experience until design operating conditions are reached.

The recommended number of heating devices or segments should not exceed five on a single protective device.

6.9 Earthing requirements

Earthing requirements include the following:

- a) The electrically conductive covering of the trace heater shall be connected to the earthing system to provide for an effective earth path.
- b) In applications where the primary earth path is dependent on the electrically conductive covering, the chemical resistance of the material shall be considered if exposure to corrosive vapours or liquids might occur.
- c) Stainless steel type braids and sheaths typically have high resistance and may not provide effective earth paths. Consideration should be given to alternative earthing means or supplemental earthing protection.

6.10 Earth-fault protection of equipment

The trace heater branch circuit protection should be capable of interrupting high-impedance earth faults as well as short-circuit faults (see IEC/IEEE 60079-30-1). This shall be accomplished by an earth-fault protective device with a nominal 30 mA trip rating or a controller with earth-fault interruption capability for use in conjunction with suitable circuit protection. The preferred trip level is nominally 30 mA or 30 mA above any inherent capacitive leakage characteristic of the trace heater as specified by the trace heating supplier.

Where conditions of maintenance and supervision help ensure that only qualified persons service the installed systems, and continued circuit operation is necessary for the safe operation of the equipment or processes, earth-fault detection without interruption is acceptable if alarmed in a manner to assure an acknowledged response.

6.11 Start-up at minimum ambient temperatures

The rating and characteristics of current protective devices should be appropriate for the trace heating systems where minimum ambient start-up conditions may occur. Refer to the trace heating supplier's instructions for additional details and recommendations in these cases.

6.12 Long trace heater runs

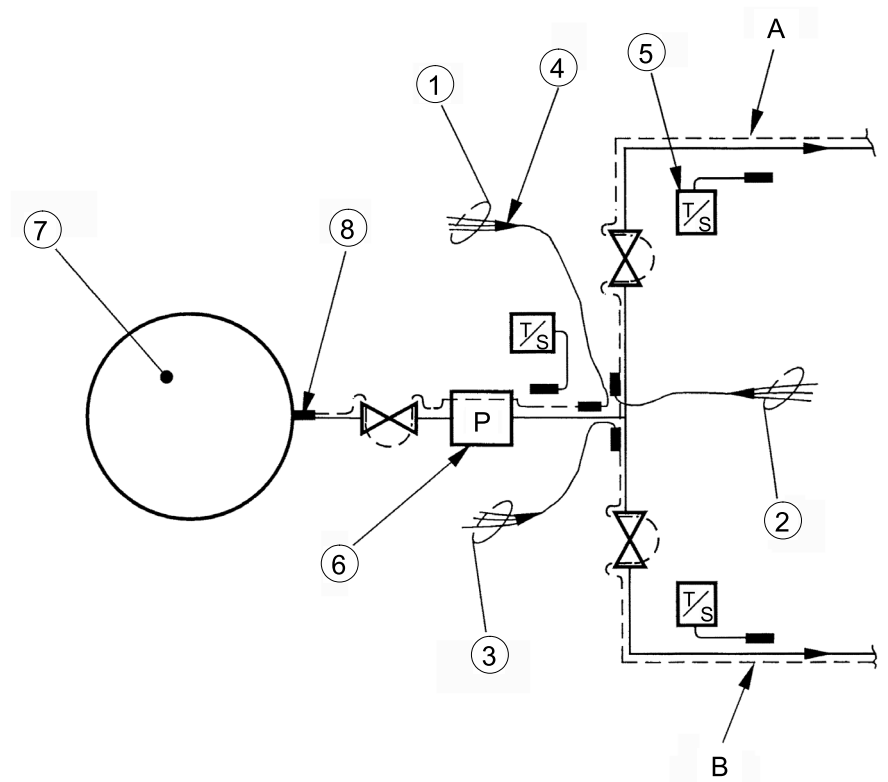
When parallel circuit trace heaters are used in long runs, the power density at the end of the run can be less than the power density at the beginning due to voltage drop. The temperature sensor should be located away from the trace heater and the power supply connection, as well as away from heat sinks.

6.13 Flow pattern analysis

Where critical temperature control is required, all possible flow conditions in the piping network shall be considered in determining the trace heating circuit segments. This is illustrated by the example shown in Figure 3. All three trace heating circuits with separate controls are necessary to maintain the piping system at its desired maintenance temperature. When the heated product flows from the tank through pipe A, circuit no. 1 and circuit no. 2 are de-energized, and circuit no. 3, which is heating the non-flowing line, remains energized. If all three circuits are combined into one, using only one control, the non-flowing line A or B is de-energized and drops below the desired maintenance temperature.

A bypass around a control valve is another common occurrence where additional circuits are needed, as shown in Figure 4.

These are two examples of piping systems for which the circuit design needs close attention. Dead legs and manifold systems require particular arrangements of the trace heating devices and their associated controls.

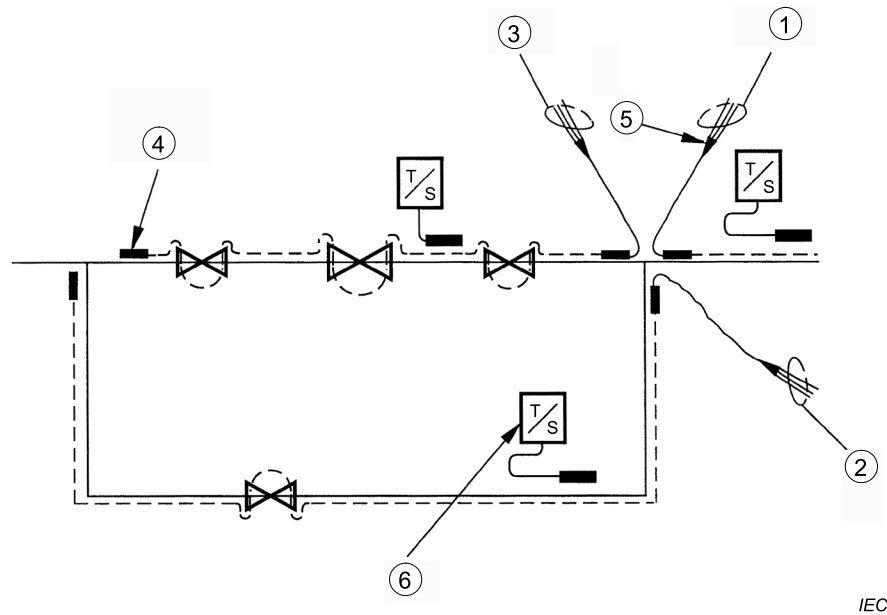


IEC

Key

- | | |
|------------------------|-----------------------|
| A pipe A | B pipe B |
| 1 circuit no. 1 | 5 temperature sensor |
| 2 circuit no. 2 | 6 pump |
| 3 circuit no. 3 | 7 heated tank |
| 4 cold end termination | 8 hot end termination |

Figure 3 – Flow pattern analysis example



Key

- | | |
|-----------------|------------------------|
| 1 circuit no. 1 | 4 hot end termination |
| 2 circuit no. 2 | 5 cold end termination |
| 3 circuit no. 3 | 6 temperature sensor |

Figure 4 – Bypass example

6.14 Dead-leg control technique

This is a technique that is sometimes used for the temperature control of very complex piping networks and manifold systems. It can also be used where the total number of temperature controllers is to be kept to a minimum at the expense of some energy savings. The technique consists of locating or fabricating a section of pipe that:

- has static flow condition at all times;
- has the same heat loss as the other piping to be controlled.

Then, regardless of flow conditions, all sections are heated. All sections that have static flow conditions at the same time then have the proper amount of heat required as the ambient temperature varies. Pipe sections that have flow may be heated unnecessarily. The merits of the dead-leg approach lie mainly in the trade-off in energy savings against savings on initial costs. This technique might not be applicable with temperature sensitive products.

Care should be taken that, first, the dead-leg section for control is long enough that its temperature is not affected by flow in the adjoining piping, and second, that the temperature sensor is located on the portion that is thermally independent of flow conditions.

6.15 Chimney effect

Due to heat transfer effects, a substantial temperature difference from the bottom to the top of long vertical runs may occur. Where close temperature control is needed, two or more control circuits may be needed. The maximum control circuit length for a long, vertical run depends on the maintenance temperature tolerance and the fluid characteristics inside the pipe. The chimney effect may be reduced by utilizing tight fitting insulation or by installing air stops of soft insulation between the pipe and insulation at one to two metre intervals of the vertical rise.

6.16 Safety shower and eyewash station design requirements

The application of heating devices to safety showers and eyewash stations involves unique concerns in addition to the general practices associated with industrial pipe heating applications. Because these are applied as part of overall safety systems for personnel, attention to water temperature and reliability of operation is important. In addition, these applications require more precise temperature control to maintain a narrow range of water temperatures, unlike most pipe heating systems.

Trace heating for emergency eyewash units, safety showers, and associated supply piping shall be correctly designed to prevent freezing or to maintain a tempered water system. For safety of personnel, the water shall not exceed a maximum temperature as determined by local codes or standards (typically 16 °C to 35 °C). Specific maintenance temperatures may be required for emergency eye wash units and safety showers in some applications. Items to be considered include the following:

- a) Prevention of scalding during use requires attention to design factors beyond normal consideration.
- b) These same design considerations shall be applied to the supply piping for the stations.
- c) It is recommended to minimize or eliminate the safety factor applied to heat loss calculations to limit the possible maximum run away pipe temperature.
- d) Tempered water systems should be designed to maintain in the range of 16 °C to 35 °C.
- e) Tempered water systems are Type III processes for purposes of temperature control (see 4.3).
- f) High-limit temperature control with high-limit alarm is also required.
- g) Recommended monitoring functions include low-temperature alarm, loss-of-voltage alarm, and year-around periodic energization of the heater to verify circuit integrity.
- h) Consider control sensor location to minimize the effects of hot spots in supply piping resulting from long vertical runs and from solar gain in outdoor overhead piping.
- i) Long vertical runs or stretches of piping with high solar gain may require the use of separate circuits to prevent wide ranges in water temperature over the entire length of the supply system.
- j) Where possible, terminations and controls should be located outside the spray pattern of the shower head.
- k) Tempered water designs should have some method to deal with algae and bacterial growth, typically through periodic flushing or recirculation.
- l) In extremely cold climates, consideration has to be given to hypothermia of washed personnel (possible use of an enclosed wash station) and to the prevention of discharged water freezing and creating a hazard in the discharge area.

7 Control and monitoring

7.1 General

A control and monitoring system shall be used which meets the application requirements of this standard that is appropriate to the application considerations for various process types according to the degree of criticality and process temperature accuracy, as outlined in 4.3.

Control and monitoring equipment may provide control of temperature such that any high-limit temperatures are not exceeded, monitoring of the trace heater circuit for faults, over-current protection, and residual current protection and isolation. It is essential that any additional requirements for particular applications, specified by the trace heating system designer for the system, also fully meet operational and safety requirements.

Where possible, the temperature control and/or limiter device should be located outside of the explosive atmosphere. In case this is not possible, the protective device shall be suitably certified for use in these areas.

7.2 Mechanical controllers

Mechanical controllers, such as thermostats, utilize two alternative principles: a bimetallic element or the expansion of a fluid confined within a bulb or a bulb and capillary. Changes in temperature result in positional displacement actuating electrical contacts that make or break the circuit.

Mechanical controllers are rugged; however, the short sensing element prevents remote panel mounting, and field calibration is cumbersome. Thermostats are mounted in the field.

Selection of the temperature sensor used with a mechanical controller shall take account of the maximum temperature rating of the sensor and its component parts, and any corrosive conditions to which it may be subjected.

Field located capillary and bimetallic type thermostats shall be provided with a type of protection suitable for use in the explosive atmosphere classification appropriate to the installation.

7.3 Electronic controllers

The sensors of electronic controllers typically comprise resistance temperature detectors (RTDs), platinum resistance thermometers (PRTs), thermistors, thermocouples (T/Cs), or other temperature sensing devices. The controllers can be located several hundred metres away from the sensor and are often located in the control and distribution panel for ease of operator and maintenance access.

These controllers electronically process the sensor signal in order to switch an electro-mechanical relay or solid-state device for on-off or phase control.

7.4 Application suitability

Freeze protection systems having a Type I (according to 4.3.1) process temperature accuracy requirement may only require a simple ambient air sensing control system. For improved energy efficiency and for process temperature accuracy Types II or III (according to 4.3.2 and 4.3.3 respectively), alternative or additional air-sensing or pipe-sensing controls should be considered.

Most process temperature applications are Types II or III, requiring sensing of pipe temperature and are often provided with at least a mechanical thermostat.

For critical applications and/or where temperatures are to be controlled within a narrow band (Type III), alarm functions such as annunciation of high and low process temperature and trace heating circuit failure may be required. When conditions or the job specification require it, electronic controls should be used. Systems are often provided with continuity, earth fault, and system diagnostic alarms and high-limit temperature switching. Depending on the system requirements, high-limit signals may be configured to operate an alarm and/or to operate the circuit protection device.

7.5 Location of controllers

Electronic controllers are often grouped in a common cabinet that shall conform to the requirements of the explosive atmosphere classification, if applicable. Consideration should be given to grouping the controllers outside the explosive atmosphere. Where possible, temperature controllers should be sited to allow free and easy access for convenient maintenance and calibration.

7.6 Location of sensors

The number and location of sensors are determined by the requirements of the process design criteria.

- Sensors should be positioned at points that are representative of the maintain temperature.
- Where two or more trace heating cables meet or join, sensors should be mounted 1 m to 1,5 m from the junction.
- If a trace heater circuit includes both piping and in-line heat sinks or heat sources, the sensor should be located on a section of pipe in the system approximately 1 m to 1,5 m from the in-line heat sinks or heat sources.
- Where a pipeline heating circuit runs through areas with different ambient temperatures (such as inside and outside a heated building), two sensors and associated controls may be required to control pipeline temperatures properly.
- In complex piping systems, the material flow patterns shall be evaluated for all possible circumstances before selecting the sensor location. Detailed information on this evaluation is given in 6.13 and 6.14.
- The temperature sensor for control should be located to avoid direct temperature effects from the trace heater. The sensor should be securely mounted to insure good thermal contact with the workpiece.
- The temperature sensitivity of certain process materials and certain types of piping materials may warrant both a control and high-limit temperature device. The control sensor should be located at least 90° around the circumference from the trace heater. The high-limit sensor may be located immediately adjacent to the trace heater with a set point at the material or system maximum allowable temperature, minus a safety margin.
- When using a high-limit temperature sensor to limit the sheath temperature in explosive atmospheres, the sensor may be located directly on the trace heater, or may be located away from the trace heater so as not to act as a heat sink. If located away from the trace heater, the set point needs to be lower than the specified high temperature limit setting (refer to clause 8.4.4.4) to compensate for the calculated differential between the pipe temperature and the sheath temperature.

7.7 Alarm considerations

7.7.1 General

The primary function of an alarm circuit is to alert operating personnel that the trace heating system may be operating outside the design limits and shall therefore be checked for possible corrective action. The type and function of the various alarm systems depend on the requirements of the process (see 4.3 and 4.4). Any, or all, of the alarms may be incorporated in data-logging equipment. Characteristics of the most common devices are listed in 7.7.2 to 7.7.4

7.7.2 Trace heating circuit alarm

A trace heating circuit alarm is used to detect loss of current, voltage, or continuity of the trace heating circuit, and includes (but is not limited to) the following:

- a) a current-sensing device which monitors the trace heater current and signals an alarm if the current drops below a pre-set minimum whilst the temperature switch is closed;
- b) a voltage-sensitive device, which monitors voltage at the end of the trace heater (usually a parallel-type heating cable) or monitors voltage on a return wire installed within the trace heater;
- c) resistance-or continuity-sensing devices which monitor the trace heating circuit when the system is de-energized. Usually, a low-voltage signal or pulse is transmitted into the trace heater and monitored.

7.7.3 Temperature alarms

Temperature alarms are used to perform the following functions.

- a) Low-temperature alarm: this indicates that the piping system and hence the process material temperature has fallen below a pre-set minimum and that subsequent cooling may be beyond acceptable operating design criteria. The alarm is incorporated with a temperature controller or fitted as a separate device.
- b) High-temperature alarm: this indicates that the piping system and hence process material temperature has exceeded a pre-set maximum temperature and that subsequent heating may be beyond acceptable operating design criteria. As with the low-temperature alarm, the alarm is incorporated with a temperature controller or fitted as a separate device.

7.7.4 Other alarms

Other available alarms include (but are not limited to) the following devices.

- a) Auxiliary contact alarm: the alarm is used to indicate when a contactor is closed and power is being supplied to the trace heating circuit. It provides information to the operator to confirm proper operation of the contactor, but does not assure proper operation of the trace heating circuit if a secondary contactor is open or if the trace heater has lost continuity.
- b) Residual current protective devices: devices with a single, selected current trip level are also available with alarm contacts. These devices monitor the electrical circuit earth-leakage current. If the total exceeds the selected device's trip current, the device trips, indicating a failure and interruption of the power to the circuit. Additionally, these monitoring devices are available with an alarm-only function.
- c) Switch-actuated alarm: the alarm is usually initiated by an auxiliary contact on the temperature controller.
- d) Current-sensing apparatus: the apparatus consists of a temperature control bypass switch and an ammeter; or current-sensitive relays and alarms.
- e) Diagnostic alarm: this alarm is initiated by a diagnostic circuit within the electronic controller, signalling failure of an internal control, of a data processing logic circuit or of temperature sensing devices.
- f) Disconnect device alarm: This alarm is activated by an auxiliary contact on the disconnect device to indicate that the disconnect device is open.

7.7.5 Integrated control

Trace heating system control and alarm circuitry may need to be integrated with a central (master) control and monitoring system. Particular consideration shall be given in the selection of equipment that is compatible in both the control and supervisory functions to assure successful and reliable data transfer.

8 Recommendations for installation

8.1 General

Each electric trace heating system is designed to meet the requirements of the particular process and plant. Because the system comprises a number of components integrated at the site, it is necessary to assure that the plant parameters on which the design is based are still valid when the trace heating system is installed, and also that the components are installed correctly. Appropriate testing and maintenance are essential in order to assure satisfactory performance and safety.

8.2 Preparatory work

8.2.1 General

All preparation should be conducted in accordance with the design documentation, and every item listed in the following should be checked after completion.

8.2.2 Scheduling and coordination

Installation of the trace heating system should not begin until all piping runs and pieces of equipment have been pressure-tested and all related instrumentation has been installed. The workpiece surface on which the trace heater is to be installed should be free from rust, grease, oil, etc. Any sharp protrusions such as weld splatter, cement splash, etc., should be removed. All coatings or finishes applied to the heated surfaces shall be suitable for the intended duty. The installation of the trace heating system should be coordinated with the workpiece, thermal insulation and instrumentation work in order to assure a scheduled completion date. Scheduling for the installation of thermal insulation should not occur until the electrical trace heating has been completely installed and tested.

8.2.3 Confirmation of equipment

The equipment on site to be traced should be verified such that the length of piping and the number of vessels, valves, flanges and components agree with the design drawings. The amount of trace heating depends on the numbers of these items. Whenever a change is made to the equipment to be traced, the schedule of trace heating materials shall be reviewed.

8.2.4 Receiving materials

Upon receipt of trace heating system components, a general inspection should be conducted including a confirmation of the correct type and quantities of materials and documentation. All trace heaters should be checked to verify catalogue type, product and package markings, power rating, voltage rating, quantity and special characteristics. In addition, receipt of installation instructions and the certificate of conformity or declaration of conformity from a notified body, as required, should be verified.

8.2.5 Warehousing and handling

Materials should be stored in protected, dry areas. Materials are to be released only as required on the jobsite, so as to avoid any unnecessary handling and inadvertent damage.

8.2.6 Personnel aspects

Persons involved in the installation and testing of electric trace heating systems should be suitably trained in all special techniques required. Installation should be carried out under the supervision of a qualified electrician who has undergone supplementary training in electric trace heating systems for use in explosive atmospheres. Only specially trained personnel shall carry out especially critical work, such as the installation of connections and terminations.

8.3 Installation of trace heating circuits

8.3.1 Coordination and equipment verification

The workpiece drawings and design data of the trace heating circuits should be available for coordinating the installation work. Prior to installation, the as-built piping and other equipment should be checked against the workpiece drawings. Any changes compared with the drawings may necessitate modification of the trace heaters. The installation of the trace heating system should be coordinated with the piping, thermal insulation and instrument disciplines to assure a scheduled completion date.

The supplier of the trace heating system is required to provide specific instructions for the trace heaters and the various types of system components. Instructions for components and trace heaters suitable and intended for unclassified and explosive atmospheres use shall be clearly identified.

8.3.2 Pre-installation testing and design verification

The following tests shall be performed and documented on an installation checklist (see Table B.1). A list of typical items to be reviewed is included in Table 1. The installation conditions shall be reviewed for each circuit design, and the circuit designer shall be notified in the event of any discrepancies.

- a) Trace heaters shall be visually checked for damage. Continuity should be measured in accordance with Table B.1, and insulation resistance should be measured in accordance with 8.3.4. The results should be recorded on the installation checklist.
- b) Individual controls shall be tested to assure correct calibration including, but not limited to, set points, operating temperature range and span.
- c) Vendor fabricated and assembled control panels shall include documentation certifying that all wiring, layout and functions are correct and have been tested. Upon receipt of the control panels at the work site, a general inspection shall be made to confirm also that no damage has occurred in transit.

8.3.3 Visual examination

The trace heaters shall be completely free of physical damage. Connections preassembled at the factory shall be sufficiently rugged to withstand normally expected conditions during installation.

8.3.4 Insulation resistance test

Insulation resistance shall be measured from trace heater conductors to the electrically conductive covering with a minimum 500 V d.c. test voltage. However, it is strongly recommended that higher test voltages are used – mineral insulated trace heaters should be tested at, but not exceeding, 1 000 V d.c., and polymeric insulated trace heaters should be tested at 2 500 V d.c.. The measured insulation resistance shall not be less than 20 MΩ.

8.3.5 Component substitution

Component substitution includes the following limitations:

- a) Components specifically listed in the supplier's installation or maintenance instructions shall not be substituted with similar parts unless the components are part of the certification.
- b) Other components specified in the supplier's installation or maintenance instructions may be substituted with any suitably rated component.
- c) Components that are part of a wiring system that supplies power to the trace heater may be substituted with any suitably rated component acceptable to the authority having jurisdiction.

8.3.6 Location of power supply

The power supply location shall be determined and specified prior to installation of the trace heaters. The junction boxes shall be mounted in such a way that the trace heater cannot suffer damage between the point at which it emerges from the insulation and the point of entry into the junction box.

Table 1 – Pre-installation checks

Items to be checked		Remarks
1	Is the workpiece fully erected and tested and all temporary supports removed? Is the surface to be heated free from sharp edges, weld spatter and rough surfaces?	Any welding or pressure testing after the installation of a trace heater could damage the device See 8.2.2, 8.3.7.1 and 8.4.1
2	Is the surface upon which the trace heater is to be applied metallic (thermally conductive) or non-metallic?	If the surface is of polished stainless steel, very thin-walled pipe or non-metallic of any kind, special precautions may be necessary
3	Do the items to be heated correspond in size, position, etc. with the design?	It is sometimes difficult to be sure that the correct pipe is being heated. A suitable line numbering system may be of assistance
4	Has it been specified that metallic foil be installed before the application of the trace heater?	This may be used to aid heat distribution
5	Has it been specified that metallic foil be installed after the application of the trace heater?	This may be used to prevent insulation from surrounding the trace heater or to aid heat distribution
6	Can flow of product under normal or abnormal conditions reach temperatures greater than those that the trace heater can withstand?	This would normally be covered in the design stage; however, further discussion with staff at the plant may show that incorrect or out-of-date information has been used
7	Is the trace heating system most recent documentation (working drawings, designs, and instructions) available?	No change shall be contemplated without reviewing the trace heating system documentation, as design calculations may need to be repeated to assure safe operation
8	Can pipes or surfaces expand and contract so as to cause stress on any part of the trace heating system installation?	In this case precautions are necessary to avoid damage
9	Can sensors of temperature controllers be affected by external influences?	An adjacent heating circuit could affect the sensor
10	Is the trace heater to be spiralled or zig-zagged onto the workpiece, according to the design?	Check design loading per unit length of pipe (or surface area) to determine if spiral or zig-zag application is necessary
11	Are cold leads, when fitted, suitable for contact with the heated surface?	If the cold lead is to be buried under the insulation, it has to be able to withstand the temperature
12	Is the pipework hung from a pipe rack?	In this case, special precautions are required to assure the weatherproofing of the insulation at points of suspension
13	Does pipework have its full complement of supports?	The addition of intermediate supports at a later stage could damage the heating system
14	Are sample lines/bleed lines, etc. at the plant but not on drawings?	These could obstruct or prevent the fitting of the trace heater, and a review of the trace heating system documentation may be necessary
15	Are other parameters used in the design of the equipment, such as pipe supports, specified by the design documentation?	More or less lengths of trace heater may be required than was called for in the design. This may require for redesign of the circuit.
16	Are the trace heaters, controllers, junction boxes, switches, cable glands, etc., suitable for the explosive atmosphere classification and the environmental conditions and are they protected as necessary against corrosion and the ingress of liquids and particulate matter?	If the trace heater design does not meet the intended application, the circuit shall be redesigned and the system documentation shall be updated.

8.3.7 Installation of trace heaters

8.3.7.1 General

Trace heaters should be attached to the piping and equipment in accordance with the supplier's instructions, on a clean, smooth portion of the equipment. Special care should be

taken at flanges, valves and other fittings to orient trace heaters in such a manner as to avoid damage from sharp or jagged surfaces, as well as damage due to impact, abrasion or vibration. A check should be made to assure that the trace heater, termination(s) and cold lead(s) can accommodate movement and vibration of the piping and equipment.

The installer should understand the importance of the tracing system to provide uniform heating of the piping and other equipment, noting that equipment with greater mass or heat sinks require additional tracing. A trace heater should be installed to provide as intimate a contact as is reasonably possible to the surface to be heated. Where such contact is not possible, such as on valves, a suitable heat conductive covering of temperature rated metal foil or other heat conducting materials may be used.

The trace heater should not be folded, twisted, or allowed to overlap, cross or touch itself unless this is specifically permitted in the supplier's instructions. Attention should be given to the supplier's minimum bending radii.

In the installation of trace heating systems, only genuine components may be used. Otherwise the system will not comply with the certificate.

Site-fabricated trace heaters are permissible. See 6.4.2.

8.3.7.2 Straight tracing runs on pipe

Multiple straight trace heaters should be equally spaced around the circumference of the pipe. Additional lengths of trace heater should be provided to compensate for the additional heat losses at pipe supports, hangers, anchors, etc.

8.3.7.3 Spiral tracing runs on pipe

Spiral pitch marking should be made on the pipe and equipment before applying the trace heater in a uniform spiral, starting at the power supply point and maintaining slight tension in the trace heater as it is applied. In no circumstances should the spacing be less than the minimum declared by the supplier.

Spiral tracing runs should be applied in such a way that valves, etc., can be easily removed or replaced. If excess or insufficient trace heater exists at the end of the section to be heated, the spiral pitch should be shortened or lengthened to retain a uniform spiral generally in accordance with the design.

NOTE While spiral tracing may be more convenient on short runs or piping and equipment, multiple straight tracing runs are often preferred for ease of installation and maintenance.

8.3.7.4 Inline equipment such as valves

Additional lengths of trace heater should be allowed in the design to compensate for the additional heat losses at valves, flanges, strainers, pumps, etc. These lengths should be applied in accordance with the supplier's instructions.

Trace heating should be installed in such a way that it can be removed to allow replacement of seals (such as gaskets at flange joints) and servicing of the inline equipment (such as valves and filters for strainers) without damage. Where permitted by IEC 60079-15 for non-sparking plugs and sockets, consideration should be given to installing quick disconnect devices on one or both sides of high maintenance inline equipment. Where trace heaters cross possible sources of leaks, for example, flanges, they should be positioned to minimize contact with the leaking medium.

8.3.7.5 Fixing and banding

Fixing materials should be suitable for the maximum exposure temperature and the environmental conditions. For straight tracing runs, they should be located at intervals not exceeding 300 mm, and for spiral tracing runs, not exceeding 2000 mm. Additional fixings should be applied at bends, flanges and other obstructions.

Metal bands or tie wire as recommended by the manufacturer should only be used to fix solid metal sheath trace heaters to piping and equipment, and tightened to maintain contact with the surface to be heated. Overtightening may damage the trace heater.

8.3.7.6 Trace heaters insulation resistance test

The test procedure from 8.3.4 shall be conducted on all trace heaters after installation. The results shall be recorded on an installation record (see Table B.1).

8.3.8 Connections and terminations

8.3.8.1 General

It is essential that all types of trace heaters are terminated correctly. Connections and terminations completed at the worksite shall follow the supplier's instructions explicitly. Factory terminated equipment should be inspected to assure that such terminations are complete, properly tagged and/or marked in conformity with IEC/IEEE 60079-30-1. It is important for the installer to review the certifications, the temperature ratings of the connections and terminations, and their suitability for the operating conditions.

Series resistance trace heaters intended for site termination should be checked before installation to assure that the installed lengths correspond to the design length and loading. When mineral insulated trace heaters are terminated at the job site, the cut ends should be sealed immediately to prevent any moisture ingress. For parallel circuit trace heaters, the total circuit length should not exceed the supplier's recommendations.

8.3.8.2 Connection kits

The connection kits of a trace heater should be securely fitted in accordance with the supplier's instructions, protected to prevent physical damage, and positioned to prevent the ingress of water or other contaminants that could adversely affect its use or suitability.

8.3.8.3 Junction boxes

Trace heating circuits shall be connected into boxes that are certified for the appropriate methods of protection, and that have suitable ingress protection. Junction boxes should be located as closely as possible to the trace heater exit point while allowing for any pipe expansion. Junction box lids should not be left open at any time.

8.3.8.4 Cold leads

Checks should be made to assure that the joints are waterproof, where applicable, and the bond to earth is valid. Cold leads, if used, should emerge from the surrounding thermal insulation in such a way that ingress of water or other contaminants is not possible. Cold leads should be protected where they exit through thermal insulation.

Cold leads shall be fitted or modified on site only in strict accordance with the supplier's instructions and conditions of use as specified by the certifications. Where cold leads have been jointed to metallic sheathed trace heaters by means of soldering or brazing, neither the trace heaters nor the cold lead should be bent near a brazed joint.

If permitted for the type of equipment, metal bands or wire as recommended by the manufacturer should be used on both sides of cold lead joints to prevent movement when fastening to pipe or equipment .

8.3.8.5 Trace heater entries and glands

Seals and/or glands supplied with trace heaters shall be certified. Glands shall be fully screwed into the box and compression applied to provide a complete seal. Unused entries shall be blanked off with suitable plugs. If unthreaded entries are used, the entry size shall provide sufficient clearance for the threaded section of the gland, with the gland backnut tightened against a compressible sealing washer to seal the gland in the enclosure.

Conductor terminations (8.3.8.9) should not be completed until after all other connections and the end termination have been assembled and the circuit insulation resistance test (8.3.8.8) is conducted.

8.3.8.6 Jointing, splicing and modifications

Jointing, splicing and modifications to the trace heater shall be carried out on site only in strict accordance with the supplier's instructions. Any such work on trace heaters in explosive atmospheres can invalidate the explosive atmospheres certification. This applies particularly to any modifications to trace heaters where any change in unit length alters the power density of the trace heater and affect the sheath temperature. Modifications shall be recorded in the system documentation.

8.3.8.7 End terminations

The end termination of a trace heater shall be securely fitted in accordance with the supplier's instructions and protected to avoid mechanical damage and the ingress of water or other contaminants that may adversely affect its use or suitability.

8.3.8.8 Trace heater circuit insulation resistance test

The test procedure from 8.3.4 shall be conducted on all trace heater circuits after installation. The results shall be recorded on an installation record (see Table B.1).

8.3.8.9 Conductor terminations

Terminals shall be of sufficient size and rating to accept the conductors, which may be solid or stranded wires or foils. Care should be taken in stripping back insulation to avoid damaging the conductors.

Crimp or compression type connectors and ferrules shall be of the correct size and of an approved type for the conductor concerned. Compression tools shall be suitable for the specific types of fittings and be in good condition.

Trace heaters that have been installed and not terminated shall be sealed to prevent ingress of moisture and shall be protected from damage pending completion of the termination.

8.3.8.10 Preparation of documentation

The type, length and electrical data of each trace heater shall be reviewed and recorded in the final documentation covering installation. The connection points and other system parameters shall be recorded in the installation documentation including the piping and instrumentation diagrams if necessary.

8.4 Installation of control and monitoring equipment

8.4.1 General

The installer is usually responsible for fixing the control and monitoring and distribution panels. These provide over-current and earth-leakage protection as well as means of isolation. Some form of temperature control or limitation is usually provided to assure safe temperatures or for energy efficiency purposes.

8.4.2 Verification of equipment suitability

The selected controllers, thermostats, sensors, and related devices shall meet the requirements of the overall system with regard to the service temperature, the IP (ingress protection) rating, and the method of protection. The certification of trace heating systems may prescribe the use of specific components. In these cases only parts specified by the manufacturer shall be used.

8.4.3 Temperature controller and monitoring devices

The sensors of the temperature controllers can be applied to the pipe or equipment, or can measure the medium temperature directly when used for Type II or Type III control, or can measure the ambient temperature when used for Type I control. The sensors are typically RTDs, capillary tube thermostats, and thermocouples.

Where sensors are mounted on surfaces, effective thermal coupling is essential. The diameter and length of the sensors can affect the temperature measurement.

Water and corrosive vapour intrusion can cause failure of temperature controllers. The cover or lid of a controller housing should be closed before, during and after installation, except when required for access.

8.4.4 Sensor considerations

8.4.4.1 General sensor installation

The sensor should be installed and positioned in accordance with the supplier's instructions. The control sensor should not be situated in areas of external radiant heat, solar gain, process heat discharge or close to a heated building. Care should be exercised to assure that the sensor can sense appropriate temperature conditions within a trace heated zone and away from the end of a pipe or a pipe support. Ambient temperature-sensing elements should be sited to not be exposed to direct sunlight or other sources of heat.

The sensor should be strapped in good thermal contact with the pipe or equipment and protected so that thermal insulation cannot be trapped between the sensor and the heated surface. Care should be taken not to damage the capillary tube, thermocouple or RTD leads, or to distort the sensor and thereby cause calibration error.

Where direct medium temperature sensing is required, the sensor should be located in thermowells at suitable positions, for example above potential sludge levels in vessels.

Excess capillary tube may be run under the thermal insulation unless the overall length exceeds 1 m, in which case the volume of the capillary may be such as to affect the calibration adversely.

Care should be taken to assure that the capillary tube, thermocouple or RTD leads emerge from the thermal insulation in a manner that does not allow the ingress of moisture.

In many cases the sensor location is defined during the system design phase. Considerations for sensor placement are defined in 7.6. The following sections describe installation considerations for specific installation methods.

8.4.4.2 Sensor installation for temperature controller

The sensor for the temperature controller is installed onto the surface of the pipe or equipment in a position that provides a temperature representative of the overall circuit. As illustrated in Figure 5, the sensor shall be positioned so as not to be influenced by the temperature of the trace heater, or other factors such as heat sinks and solar gain.

8.4.4.3 Sensor installation for temperature limiting device

The sensor for the temperature limiting controller is installed onto the surface of the pipe or equipment in a position that provides a temperature representative of the overall circuit. In order to assure that the safety temperature controller can accurately react to the maximum trace heater sheath temperature, particular attention shall be paid to the location, method of attachment and set point. This method of sensor installation is based on the known relationship between the equipment temperature and the heater sheath temperature at a given power output. Typical temperature limiting sensor installation is indicated in Figure 5.

It is important that the controller is set such that the heater sheath temperature does not exceed the high-limit temperature under worst-case conditions (e.g. voltage +10 %, tracer at upper limit of manufacturing power tolerance, heater out of contact with the pipe/equipment, maximum ambient, no external convection).

For trace heater circuits that are designed for use with voltage regulating devices, it may be necessary to install the sensors using the methods described in 8.4.4.4 and 8.4.4.5. These methods respond quickly to rapid changes in heater sheath temperatures caused by failure of the voltage control device.

8.4.4.4 Temperature limiting device with sensor on trace heater sheath

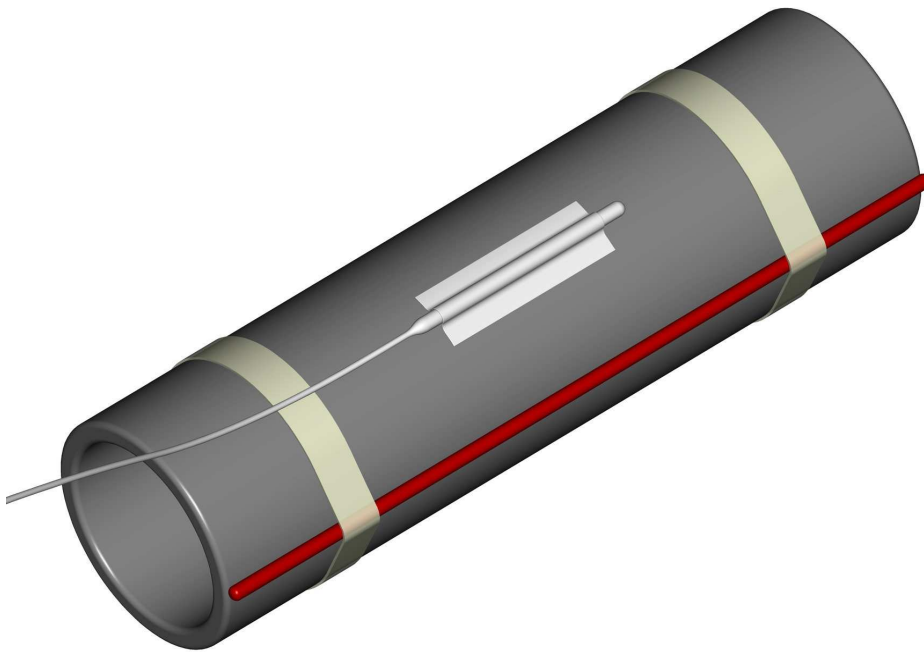
In Figure 6, the temperature sensor is mounted directly to the trace heater and the trace heater is in direct contact with the surface being heated. To insure accurate thermal coupling with the trace heater, it is necessary to install the sensor with metallic foil tape or heat transfer compound.

It is important to verify that the location is representative of the hottest point. The means to secure the sensor to the heater shall insure the sensor cannot loosen with time and temperature exposure, and that it cannot be loosened during future maintenance operations.

This method of sensor installation does not measure the hottest part of the tracer (which probably occurs at a point of no contact with the equipment). It should only be used with a controller setting that is below the high-limit temperature.

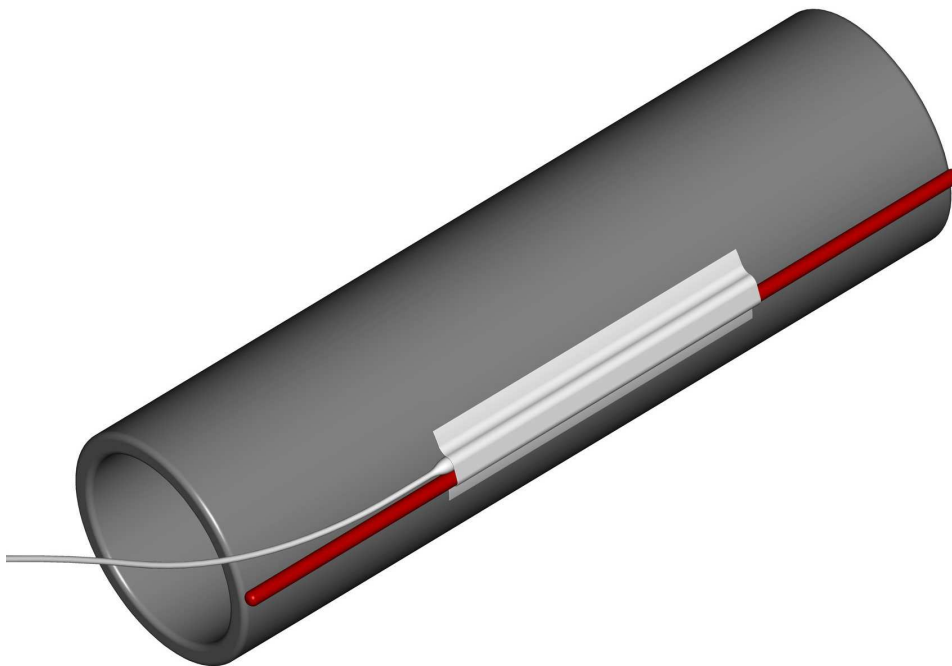
This method has the advantage that it minimizes over-temperature caused exclusively by the measuring point. In some cases, however, the measured hot spot temperature may be lower than the highest trace heater sheath temperatures at points with poor thermal coupling between the trace heater and the heated surface. This shall be taken into account in the limiter's temperature setting, just like the sensor's heat dissipation. These variances are a function of the ratio of the masses of the sensor and the trace heater (ratio of the diameters) and of the specific heating capacity (W/m).

The direct contact with the heated surface and the resulting thermal inertia may make it necessary to use an even lower setting in order to take transient phenomena (e.g. those caused by failed semiconductor contact elements) into account. In some cases it may be necessary to divide complex circuits into circuits with individual temperature limiters.



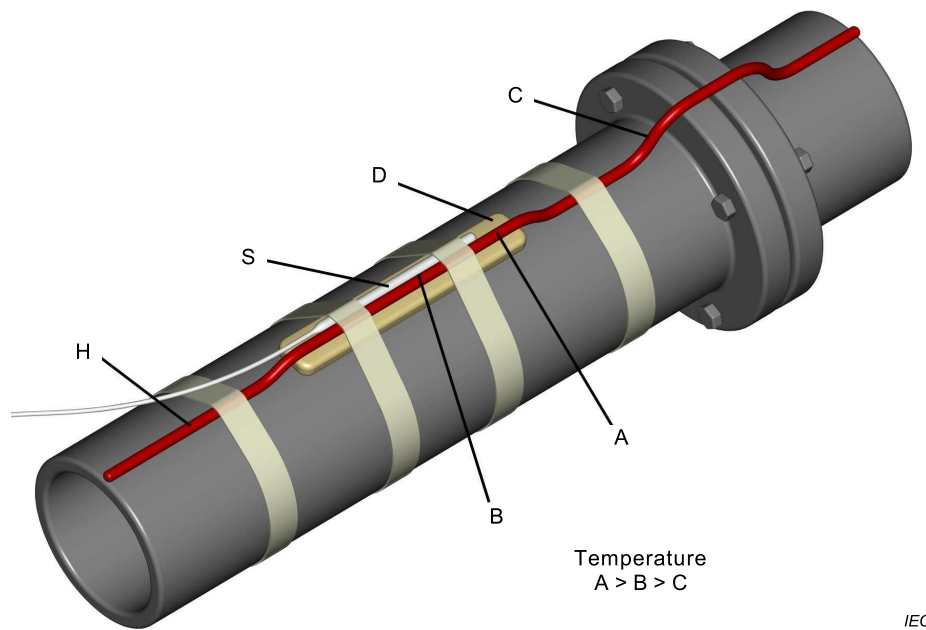
IEC

Figure 5 – Typical installation of control sensor and sensor for temperature limiting control



IEC

Figure 6 – Limiting device sensor on sheath of trace heater



Key

- H trace heater
- S temperature sensor
- A temperature of artificial hot spot
- B temperature at measurement point
- C point with poor thermal coupling (characteristic hot spot)
- D thermal insulation between trace heater and heated surface

Figure 7 – Limiting device sensor as artificial hot spot

8.4.4.5 Temperature limiting device with artificial hot spot

In Figure 7, the sensor is located to measure an artificial hot spot intending to represent the hottest point of the tracer. This may be a suitable alternative method when used with series trace heaters.

In cases where the method in 8.4.4.4 cannot reliably be utilized to coordinate to the system's worst case sheath temperature, the artificial hot spot method can provide some additional margin of safety. In this case thermal insulation is inserted between the trace heater and the surface being heated. The sensor for the temperature limiter is then installed in direct contact with the trace heater.

To assure that the temperature at the artificial hot spot is higher than the trace heater temperature at the point with poor thermal coupling (temperature C), the thermal insulation is about twice as long as the sensor. Because of the unavoidable heat dissipation of the sensor itself, the measured temperature B is indeed higher than temperature C, but lower than the actual temperature of hot spot A. This variance, which is a function of the ratio of the masses of the sensor and the trace heater (ratio of the diameters) and of the specific heating capacity (W/m), is taken into account in the limiter's temperature setting.

The advantage of this method is its extremely short reaction time to malfunctions such as controller failure, failure of the voltage control device, or overvoltage. In some cases it may be necessary to divide complex circuits into circuits with individual temperature limiters.

8.4.5 Controller operation, calibration, and access

The settings of the temperature controllers and safety temperature limiters shall be reviewed during commissioning. Depending on the setting possibilities offered by the safety temperature limiters, the limiters shall be sealed against tampering.

The temperature control device and sensor loop shall be calibrated at commissioning. The controller shall be set to the required temperature and re-calibrated from the factory setting if necessary. A function check shall be made by adjusting the temperature setting until the controller is seen to energize the trace heater.

All measured data shall be documented.

8.4.6 Necessary modifications

Trace heating systems may be specified for verification of the maximum sheath temperature. If the temperatures measured deviate from the admissible sheath temperatures or from the design figures, corrective measures shall be taken and the system design shall be re-evaluated.

8.5 Installation of thermal insulation system (see also Clause 5)

8.5.1 General

The selection and application of the thermal insulation are key elements in the installation of an electric trace heating system. The function of thermal insulation is to reduce heat losses or heat gains when the equipment normally operates above or below the ambient temperature. Consequently, problems with the insulation have a direct effect on the performance of the system as a whole. Minimization of energy dissipation reduces the running costs, improves the system characteristics and enhances the system's heating capacity. Installation of thermal insulation shall conform to all applicable national standards and local regulations.

8.5.2 Preparatory work

Precautions shall be taken to protect trace heaters from mechanical damage and moisture intrusion after they have been installed and prior to the application of thermal insulation. Before starting the installation of thermal insulation, it is recommended that site engineering establish a liaison between the trace heaters installer and the thermal insulation contractor, so that the thermal insulation is applied as soon as possible after the installation and testing of trace heaters. The installed trace heating system shall have been tested in accordance with 8.3.8.8.

The following checks and procedures should be confirmed.

- a) Verify that the type, inside diameter, and thickness agree with the values used in selection of the trace heater(s). If the insulation thickness differs from the specification, the specified design temperatures cannot be met.
- b) For thermal insulation, temporary weather protection should be provided during storage, handling and installation to avoid the risk of moisture being trapped under the weather-protective coating or jacket.

8.5.3 Installation of the thermal insulation materials

Thermal insulation should be applied to all sections of the pipe and equipment, including flanges, valves, pipe supports, bends, T-junctions, etc. If expansion joints or bellows are installed in the system, provision should be made for their thermal insulation in such a manner that it does not impair the thermal efficiency of the trace heating system.

Oversized thermal insulation may be required in order to assure that the trace heater and equipment are adequately covered. Other considerations include the following:

- a) It is essential to verify that the insulation thickness equals the specified nominal thickness at all points. Care shall be taken not to embed the trace heater in the insulation, because this could cause an increase in sheath temperature. If the specified insulation size does not fit correctly, the next greater insulation size can be used in order to accommodate the trace heater.
- b) All penetrations should be sealed to prevent the ingress of moisture. Wherever possible, cutouts should be prepared in advance and should be in the lower 180° segment of the thermal insulation. The thermal insulation should be applied in such a way that permits absolutely tight entry of the trace heaters and thermal sensors or capillary tubes.
- c) The thermal insulation should be cut and tightly fitted to avoid air gaps. Stagger segment joints on the horizontal plane to minimize convective heat loss.
- d) During application of the insulation, care shall be taken not to damage the electric trace heater. The positions of trace heaters, temperature sensors and other devices shall not be altered.
- e) The use of metal foil should be considered for covering the trace heater on valves and other irregularly shaped equipment to prevent the thermal insulation from surrounding the trace heater.
- f) Care should be taken to avoid high halide content thermal insulation materials over trace heaters with an exposed stainless steel covering.

8.5.4 Cladding

Where metal cladding is specified, particular care should be taken to assure that bare edges of metalwork cannot come into direct contact with the trace heater or its components.

Areas of greatest risk are as follows:

- a) Flanges: metalwork should be cut back and the exposed face of the thermal insulation finished with a suitable non-absorbent compound.
- b) Valves: a preformed insulating jacket should be cut overlength and carried on to adjacent pipework cladding.
- c) Pipe bends, elbows or tees: care should be taken not to force the adjacent straight pipe section of cladding into the bend and thereby risk damage to the trace heater.

Bends with rolled edge interlocking sections are preferred. A non-hardening sealer should be used between the overlapped sections of metallic weather barriers or jackets. Where rivets or self-tapping screws are used, care should be taken to assure that any drill or screw selected for use is not long enough to penetrate the thickness of the thermal insulation and damage the underlying trace heater.

Warning labels shall be affixed to the cladding at intervals not exceeding 6 m to advise that an electric trace heating system is installed beneath the thermal insulation. These labels shall also be placed on the cladding over each valve or other piece of equipment that may require periodic maintenance.

8.5.5 Field (site work) circuit insulation resistance test

The test procedure from 8.3.4 shall be conducted on all trace heater circuits after installation, with the requirement that the measured insulation resistance shall not be less than 20 MΩ, except that MI trace heaters shall not be less than 5 MΩ.

8.5.6 Visual inspection

The visual inspection shall assure that:

- a) no moisture can penetrate the insulation as a result of weathering (correct position of overlaps or lock beading);

- b) sliding connections (or the like) on weather cladding are sufficiently flexible to absorb any expansion movement;
- c) the screws selected for fastening the weather cladding are short enough to exclude any possibility of damage to trace heaters or to temperature sensors;
- d) the entry cutouts in the weather cladding for trace heaters, temperature sensors, etc., are dimensioned so as to render contact impossible. Especially in the case of branches, the cladding shall be cut sufficiently wide;
- e) the cladding joints and thermal insulation entries are properly sealed with an elastic, non-hardening sealant that is resistant to chemical attack and decay, and is dimensionally stable.

8.5.7 Documentation

The thermal insulation material and its thickness shall be documented.

8.6 Installation of distribution wiring and coordination with branch circuits

8.6.1 General

The branch circuit wiring of each trace heater circuit requires an over-current protective device. The size and type of distribution wiring, and the ratings of the branch circuit protective devices should be based on heater start-up currents and their duration at the minimum temperature the trace heater may experience.

8.6.2 Earth-fault protective device

An earth-fault protective device shall be provided as outlined in IEC/IEEE 60079-30-1.

8.6.3 Circuit protective device

It is necessary to check that:

- the protective device is sized correctly to the rated current (at initial and detailed inspections);
- the protective device is certified to all applicable national and international requirements, and to the requirements for the appropriate method of protection.

8.6.4 Tagging/Identification

Permanent tagging and identification shall be completed as follows and shall be verified for compliance with the marking requirements of IEC/IEEE 60079-30-1:

- a) branch circuit breaker;
- b) monitor and alarm apparatus;
- c) trace heater power connection;
- d) circuit number and setpoint for each temperature controller.

Marking shall be carried out in accordance with IEC/IEEE 60079-30-1 for each trace heating circuit, on the respective junction box.

8.7 Commissioning

8.7.1 Pre-commissioning check

The installation checklist given in Table B.1 should be completed and retained.

8.7.2 Functional check and final documentation

8.7.2.1 General

The trace heating system(s) shall be commissioned after the thermal insulation has been installed and the electrical distribution is completed. The trace heater commissioning record given in Table C.1 shall be completed and retained.

8.7.2.2 Functional check

The following functional checks should be conducted:

- a) Close all branch circuits and verify proper current. A temporary bypass may be required for the temperature control device.
- b) Verify that monitor or alarm circuits are operable. A bypass may be required at field contacts.
- c) Fill out the trace heater commissioning record (Table C.1) for each circuit. This shall clearly document all testing and commissioning data.
- d) Record the electrical insulation resistance values for each measurement taken according to the procedure given in 8.5.5.
- e) Record the applied voltage and resulting current after 5 min of energization, and pipe temperature if required.
- f) Verify that the alarm and monitor components operate as intended.
- g) Verify that the calibration check at the temperature controller setpoint has been performed and the controller has been set at this value.

8.7.2.3 Final documentation

Adequate and uniform documentation of the electric trace heating circuits is an essential precondition for economical maintenance of this equipment. This is especially important to facilitate rapid troubleshooting in the event of circuit problems. It also provides the basis for simpler, faster and less costly handling of any desired modifications and expansions by a specialist for electric trace heating systems.

Considering electric trace heating systems in explosive atmospheres, the form of the project documentation is specified in detail in the respective system description.

The documentation of each heating circuit of a trace heating system shall include the following elements:

- a) Design and testing documentation:
 - 1) Table of contents
 - 2) Piping layout showing the trace heating circuits and the location of power points, connections, splices, tees, end terminations, and temperature sensors for control and limitation
 - 3) For vessels: layout of the trace heating circuits
 - 4) Pipe and insulation list
 - 5) Individual circuit length of trace heaters
 - 6) Calculation and dimensioning data
 - 7) Material list
 - 8) Trace heater installation instructions
 - 9) Heater cabling plan
 - 10) Description of and installation instructions for temperature sensors
 - 11) Heater commissioning record (Table C.1)

- 12) Temperature profile measurement
- 13) Installation certificate
- b) Circuit diagrams or lists:
 - 1) Wiring and circuit diagrams or lists
 - 2) Terminal connection diagrams, switchgear with parts list
 - 3) Installation instructions
- c) Other:
 - 1) Technical descriptions and instruction manuals for the individual pieces of equipment
 - 2) Functional diagram as agreed to with the design engineer
 - 3) Certificates from a certification agency for explosive atmosphere equipment

9 Maintenance

9.1 General

It is recommended that the maintenance schedule given in Table D.1 should be undertaken at one year intervals or more frequent as required. More detailed inspection may be necessary in corrosive or other hostile environments or where there is a high risk of mechanical damage. The need for more frequent inspection should be determined by operational experience and by consultation with the user/occupier of the plant.

All maintenance activities should be recorded in a maintenance log (such as that shown in Table D.1) and retained in the system documentation.

9.2 Fault location

Specialized methods of fault location are necessary to find faults in electric trace heating systems covered by thermal insulation and metallic cladding, and advice should be sought from the electric trace heating system designer. Often faults are caused by mechanical damage, corrosion, overheating or ingress of moisture.

A brief outline of the steps that may be necessary is given below.

- a) The exact layout of the electric trace heating system should be determined from the system documentation. If this proves inadequate, an "induction" instrument operating at approximately 1 000 Hz should be used to inject a signal into the electric trace heating device and the route of the device should be followed by the audible signal obtained from the instrument.
- b) The type of fault should be determined, for instance, open-circuit or leakage-to-earth.
- c) If there is an open circuit or low resistance to earth of less than approximately 500 Ω , a pulse echo or reflection instrument can be used with a considerable degree of success. Other faults should be located using a resistance bridge-type instrument.

9.3 Fault rectification

When the fault has been located, the defective component should be replaced or repaired in accordance with the requirements of this standard. Those parts of the installation that have been disturbed should be checked in accordance with Table C.1 and recorded in accordance with Table D.1.

10 Repairs

10.1 General

After the cause of a circuit fault has been determined, the defects should be rectified by site repair or replacement. Site repair shall be carried out only if the following conditions are satisfied:

- a) It does not invalidate the certificate for certified apparatus. Any repairs to certified apparatus used in explosive atmospheres shall be strictly in accordance with any specific instructions incorporated in the certificate schedule.
- b) Design and construction characteristics of the trace heater are maintained, for example mechanical strength and water resistance.
- c) A method of repair is recommended in the trace heating system documentation using any special materials and tools.
- d) No local hazard is created in carrying out the repair.
- e) Personnel involved in repair have been trained in accordance with 8.2.6.

Subclause 8.3.5 covers the requirements for component replacement.

10.2 Practicability of repair to electric trace heaters

10.2.1 Mechanical damage

NOTE If the trace heater has not previously been energized and mechanical damage has resulted in breakdown in electrical insulation, severed conductors or ingress of moisture, repair should normally be possible.

If the fault is found only after electrical connection and the damage is confined to a small area, visual inspection of the trace heater for 1 m on either side of the fault should be undertaken to show whether the electrical insulation is affected other than at the point of mechanical damage.

10.2.2 Damage due to corrosion

NOTE Where breakdown results from corrosive action and the damage is limited to a small area, a repair should normally be possible.

If the electrical trace heater has been damaged at more than one point or the damaged area is extensive, it should be replaced.

10.2.3 Damage due to overheating

Repairs should be carried out only when damage is limited to a small area.

If a system design fault is suspected, an evaluation of the trace heater system should be conducted.

10.3 Repair techniques for electrical trace heaters

10.3.1 General

NOTE Techniques employed in the repair of electrical trace heaters vary with the type of trace heater and its manufacturer. It is therefore outside the scope of this standard to detail actual methods.

Generally, the repair of an electrical trace heater takes the form of an in-line splice or a connection via a junction box.

General procedures that should be adopted are given in this clause. However, only those methods recommended by the supplier should be adopted and only materials or tools recommended by the supplier should be used.

10.3.2 In-line splice

The removal of a damaged section of trace heater should not significantly alter the performance of the trace heater from its original design characteristics. Care should be taken to insure that an in-line splice is not subjected to stresses in operation. This may be achieved, for example, by providing an expansion loop on either side of the joint. The trace heater 150 mm on either side of the joint should not be bent when re-applied to the workpiece and good contact should be assured.

10.3.3 Connection via junction box

The repair shall not alter the performance of the trace heater circuit from its original design characteristics. Where cold leads are applied, the trace heater and repair joints should be firmly reattached to the workpiece to assure good contact. Fitting of cold leads and terminations into the junction box should be in accordance with the supplier's instructions.

10.4 Earthing

Where trace heaters are earthed by means of an electrically conductive covering, the integrity and continuity of the earth shall not be impaired by the repair.

10.5 Testing

The repaired trace heater circuit(s) shall be subjected to the tests described in 8.5.5 of this standard before re-installation.

Annex A (informative)

Example of design data record

Table A.1 gives an example of a design data record.

Table A.1 – Example of design data record

Location	System	Project Number	Reference Drawing(s)
SITE INFORMATION			
Minimum Ambient Temperature		Design Wind Speed	
Maximum Ambient Temperature		Design Safety Factor	
Installed:	Outdoors	Indoors	Severe (Corrosive) Environment
SYSTEM REQUIREMENTS			
Type I (Temperature Maintained above a minimum point)			
Type II (Process maintained within a moderate band)			
Type III (Process controlled within a narrow band)			
APPLICATION			
Pipelines	Vessels	Safety Showers	
Non-Metallic Pipe / Vessels		Pre-Traced Instrument Tubing	
Steam Condensate Lines (freeze protection)			
Allow Spiraling of tracer			
PROCESS INFORMATION			
Material in Pipe		Liquid / Gas / Vapor	
Pipe Maintenance Temperature		deg C	
Normal Process Operating Temperature		deg C	
Minimum Allowable Product Temperature		deg C	
Maximum Allowable Product temperature		deg C	
Maximum System Temperature (from process excursions, steamout, etc.)		deg C	
PIPING (VESSEL) SYSTEM			
Pipe (Vessel) Material		Schedule (Thickness)	
Special Conditions (lined pipe, etc.)			
Pipe Supports Method(s)	Direct Contact	Outside thermal insulation (preferred)	
THERMAL INSULATION SYSTEM			
Type	Thickness	K-value / Temp	
Maximum Exposure Temperature		deg C	
Soft Insulation used (Valves, pumps)		Installed Oversized	
ELECTRICAL SYSTEM			
Voltage(s) Available	Volts	Phase	Hertz
Electrical Area Classification		AIT / T-Rating	
Determining Gas/Vapor (lowest AIT)			
Approvals required		IEC	PE Stamped Drawings
SPECIAL PROCESS HEATING CONSIDERATIONS			
Use this section only for Heat Up, Melt Out or other special heating requirements.			
Special Heating Requirement	Heat Up	Melt Out	Other, describe:
Volume of fluid / solid to be heated		Flowing / Non-flowing	
Allowable time to accomplish rise in temperature / change state			
Initial material temperature		Final material temperature	
Temperature when material changes state		Pipe Material	
Specific Heat	Solid	Liquid	Vapor
Density	Solid	Liquid	Vapor
Heat of Fusion or vaporization			
Prepared by	Company	Date	
Approved by	Company	Date	
Received by	Company	Date	

Annex B (informative)

Checklist for installation requirements

Table B.1 gives an example of a pre-commissioning check and trace heater installation record.

Table B.1 – Example of pre-commissioning check and trace heater installation record

Location	System	Project number	Reference drawing(s)
Line number	Trace heater number	Area classification	Temperature classification
Panel number	Location	Circuit number	Circuit amp/voltage
Trace heater manufacturer	Trace heater model	Trace heater wattage unit length/voltage rating	
Verify certification marking:			
Megohm meter manufacturer/model		Voltage setting	Accuracy/full scale
Megohm meter date of last calibration			
Multimeter manufacturer/model		Ohm setting	Accuracy/full scale
TRACE HEATER TESTING	Test value/remarks	Date	Initials
NOTES: Continuity test on self-regulating trace heater only used for short or open circuit. Minimum acceptable insulation resistance shall be 20 MΩ when measured at the trace heater junction box. Minimum acceptable test voltage is 500 V d.c. However, 1 000 V d.c. recommended for MI, 2 500 V d.c. for polymeric trace heaters.			
1 Receipt of material on reel			
Continuity test on reel			
Insulation resistance test on reel			
2 Piping completed (approval to start trace heater installation)			
3 Trace heater installed (approval to start thermal insulation installation)			
Trace heater correctly installed on pipe, vessel or equipment			
Trace heater correctly installed at valves, pipe supports, other heat sinks			
Components correctly installed and terminated (power, tee-end seal)			
Installation agrees with manufacturer's instructions and circuit design			
4 Thermal insulation installation complete			
Continuity test			
Insulation resistance test			
SYSTEM INSPECTED			
5 Marking, tagging and identification complete (see IEC/IEEE 60079-30-1, Clause 6)			
6 Trace heater effectively earthed			
7 Temperature controls properly installed and set points verified			
8 Junction boxes all certified and closed			
9 Thermal insulation weather tight (all penetrations sealed)			
10 End seals, covered splices marked on insulation outer cladding			
11 Drawings, documentation marked as-built			
Performed by:		Company	Date
Witnessed by:		Company	Date

Accepted by:	Company	Date
Approved by:	Company	Date

Annex C (informative)

Example of trace heater commissioning record

Table C.1 gives an example of a trace heater commissioning record.

Table C.1 – Example of trace heater commissioning record

Location	System	Project number	Reference drawing(s)			
Line number	Trace heater number	Area classification	Temperature classification			
Panel number	Location	Circuit number	Circuit amp/voltage			
Trace heater manufacturer	Trace heater model	Trace heater wattage unit length/voltage rating				
Verify certification marking:						
TRACE HEATER INFORMATION						
Trace heater total design length				Trace heater total installed length		
Thermal insulation type				Thermal insulation thickness		
Workpiece maintain temperature				Maximum workpiece temperature		
TRACE HEATER TESTING (data from trace heater installation record)						
Electrical resistance (continuity) test, in ohms						
Electrical insulation resistance test, in megohms						
Test ambient temperature						
PERFORMANCE DATA	Volts a.c		Current in amperes			
	Panel	Field	Single-phase	Three-phase		
			Line	L1	L2	L3
Start-up						
After 5 min						
After 4 h						
Ambient temperature at time of test						
Workpiece temperature at beginning of test			After 4 h			
Calculated watts per unit length ($V \times A/m$)			After 4 h			
TEMPERATURE CONTROL: type						
Trace heater controller	Ambient sensing		Workpiece sensing		Temperature setpoint	
High limit controller	Type		Location		Temperature setpoint	
Heating controls calibrated						
Heating controls operation verified						
ALARMS/MONITORING: type						
Temperature	High setting		Low setting		Operation verified	
Trace heater current	High setting		Low setting		Operation verified	
Residual current			Setting		Operation verified	
Loss of voltage					Operation verified	
Other					Operation verified	
RCD PROTECTION: type						
Setting	Measured current			Tested in operation		
Performed by:				Company	Date	

Witnessed by:	Company	Date
Accepted by:	Company	Date
Approved by:	Company	Date

Annex D (informative)

Example of maintenance schedule and log record

Table D.1 gives an examples of a maintenance schedule and log record.

Table D.1 – Example of maintenance schedule and log record

Location system		System			Reference drawing(s)		
CIRCUIT INFORMATION							
Trace heater number		Circuit length			Breaker panel no.		
Power connection		Design voltage			Breaker pole(s) no.		
Tee connection		Residual current protection (type)					
Splice connection		Residual current trip setting					
Heating controller							
VISUAL							
Panel no.	Circuit no.						
	Date						
	Initial						
Thermal insulation							
Damaged insulation/ lagging							
Water seal good							
Insulation/lagging missing							
Presence of moisture							
Heating system components							
Enclosures, boxes sealed							
Presence of moisture							
Signs of corrosion							
Trace heater lead discolouration							
Heating and/or high limit controller							
Operating property							
Controller set point							
ELECTRICAL							
Insulation resistance testing (bypass controller if applicable) refer to 8.5.5 in this standard.							
Test voltage							
Megger value, MΩ							
Trace heater supply voltage							
Value at power source							
Value at field connection							
Trace heater circuit current reading							
Amps reading at 2 min to 5 min							
Amps reading after 15 min							
Earth-fault current							

Comments and actions		
Performed by:	Company	Date
Approved by:	Company	Date

Annex E (informative)

Pipe heat loss considerations – Heat loss formula and example calculations

There are many variations of the general formula [Formula (E.1)] that defines the heat loss from a pipe. Some of these variations are simplifications made for the ease of calculation, conservative (high) values or similar reasons. Other variations reflect the absence of terms that are dependent on the insulation system. The variation selected is therefore dependent on the situation at hand and on the desired goal:

$$q = \frac{T_p - T_a}{\frac{1}{\pi \times D_1 \times h_i} + \frac{\ln(D_2 / D_1)}{2 \times \pi \times K_1} + \frac{\ln(D_3 / D_2)}{2 \times \pi \times K_2} + \frac{1}{\pi \times D_3 \times h_{co}} + \frac{1}{\pi \times D_3 \times h_o}} \quad (\text{E.1})$$

where

- q is the heat loss per unit length of pipe (W/m)
- T_p is the desired maintenance temperature (°C)
- T_a is the minimum design ambient temperature (°C)
- D_1 is the inside diameter of the inner insulation layer (m)
- D_2 is the outside diameter of the inner insulation layer (m) (inside diameter of the outer insulation layer when present)
- D_3 is the outside diameter of the outer insulation layer when present (m)
- K_1 is the thermal conductivity of the inner layer of insulation evaluated at its mean temperature (W/ m °C)
- K_2 is the thermal conductivity of the outer layer of insulation, when present, evaluated at its mean temperature (W/m °C)
- h_i is the inside air contact coefficient from the pipe to the inner insulation surface when present,
- h_{co} is the inside air contact coefficient from the outer insulation surface to the weather barrier when present (W/m² °C)
- h_o is the outside air film coefficient from the weather barrier to the ambient air (W/m² °C). Typical values for this term range from 3 W/m² °C to 284 W/m² °C for low (below 50 °C) temperature applications.

Formula (E.1) includes all common resistances to heat flow that may be present. For a typical insulation system consisting of a single type of insulation, no oversizing, and a metal weather barrier, Formula (E.1) reduces to Formula (E.2):

$$q = \frac{T_p - T_a}{\frac{\ln(D_2 / D_1)}{2 \times \pi \times K} + \frac{1}{\pi \times D_2 \times h_{co}} + \frac{1}{\pi \times D_2 \times h_o}} \quad (\text{E.2})$$

When a mastic weather barrier is used, Formula (E.2) reduces to Formula (E.3):

$$q = \frac{T_p - T_a}{\frac{\ln(D_2 / D_1)}{2 \times \pi \times K} + \frac{1}{\pi \times D_2 \times h_o}} \quad (\text{E.3})$$

Finally, the $1/(\pi)(D_2)(h_o)$ term may be omitted to give a conservative (high) heat loss. Formula (E.3) then becomes Formula (E.4), which expresses heat loss per unit length:

$$q = \frac{2 \times \pi \times K \times (T_p - T_a)}{\ln(D_2 / D_1)} \quad (\text{E.4})$$

Formula (E.5) gives the total heat loss for a pipe of length L :

$$q = \frac{2 \times \pi \times K \times L \times (T_p - T_a)}{\ln(D_2 / D_1)} \quad (\text{E.5})$$

It should be noted that the film coefficients and insulation thermal conductivities are functions of temperature. Initially, the temperature at the location of each resistance is not known and should be assumed. Based on the initial temperature profile assumption, a heat loss and a new temperature profile is calculated. If agreement between the assumed temperature profile and the calculated temperature profile occurs, then a solution has been obtained. If not, then the calculated temperature profile should be used for the next assumed temperature profile and the calculation should be repeated (Figure E.1).

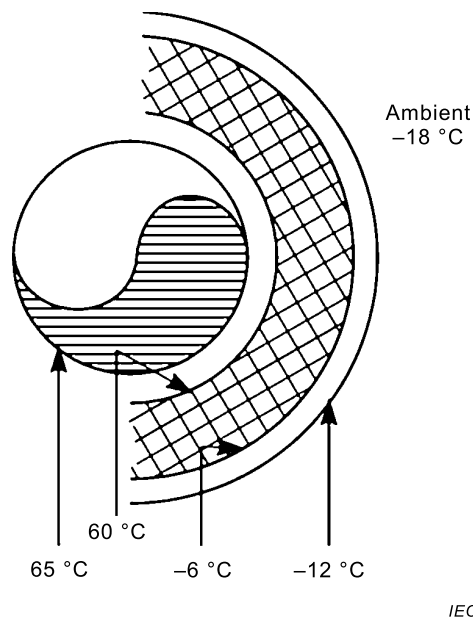


Figure E.1 – Assumed temperature gradients

Air film coefficients h_{co} , h_o , and h_i are effective values based on the combined heat losses from convection and radiation. The convective portion of these coefficients is highly dependent on the velocity of the air surrounding the surface. If the air is still, then heat loss from convection is denoted as free convection. If the air is circulated by wind, then the loss is termed as forced convection. Estimation of heat transfer coefficients from convection is based on experimental data. These data essentially relate Nusselt numbers to Rayleigh numbers for free convection (McAdams [B21]) and Nusselt numbers to Reynolds and Prandtl numbers for forced convection (Holman [B10] and McAdams [B21]). Because electric heat tracing systems are predominantly concerned with heat loss from cylindrical pipes in air at atmospheric

pressure conditions, the empirical relationships describing these heat transfer rates can be simplified as follows.

Simplified formulae for laminar free convection in air at atmospheric pressure (McAdams [B21]) are shown in Formula (E.6) and Formula (E.7):

$$h = C_1 \times \left[\frac{T_s - T_{amb}}{d} \right]^{0.25} \quad \text{for horizontal pipes} \quad (E.6)$$

$$h = C_2 \times \left[\frac{T_s - T_{amb}}{L} \right]^{0.25} \quad \text{for vertical pipes} \quad (E.7)$$

where

- h is the heat transfer coefficient from free convection (W/m² °C)
- C_1 is 1,32
- T_s is the temperature of cylindrical surface (°C)
- T_{amb} is the temperature of surrounding ambient air (°C)
- d is the diameter of cylindrical surface (m)
- C_2 is 1,42
- L is the vertical length of cylinder (m)

Simplified formulae for forced convection for a cylinder in air at atmospheric pressure (Holman [B10] and McAdams [B21]) are shown in Formula (E.8):

$$h_f = \frac{C_3 \times k_f}{d} \times \left(\frac{Vd}{\nu_f} \right)^n \times (Pr)^{1/3} \quad (E.8)$$

where

- h_f is the heat transfer coefficient from forced convection (W/m² °C)
- C_3 is 0,266, the empirical dimensionless constant
- k_f is the thermal conductivity of air evaluated at the average air film temperature (W/m² K)
- d is the diameter of cylinder (m)
- V is the wind velocity (m/s)
- ν_f is the kinematic viscosity, evaluated at the average air film temperature (m²/s)
- n is 0,805, the empirical dimensionless constant
- Pr is the Prandtl number for air evaluated at the average air film temperature (dimensionless)

NOTE C_3 and n are constant for Reynolds numbers in the range of 40 000 to 400 000, which is sufficient for most typical heat tracing applications. The Reynolds number (Re) is a dimensionless number defined as follows:

$$Re = \frac{Vd}{\nu_f}$$

where these variables are defined identically to the variables in Formula (E.8).

Heat loss from radiation from a cylinder is described by the Stefan-Boltzmann law of thermal radiation (McAdams [B21]), as shown in Formula (E.9):

$$q_r = \sigma \times \varepsilon \times A \left[(T_1 + T_0)^4 - (T_2 + T_0)^4 \right] \quad (\text{E.9})$$

where

- q_r is the heat loss of cylinder (W)
- δ is $5,669 \times 10^{-8}$ (W/m² × K⁴)
- ε is the emissivity of radiating surface (dimensionless)
- A is the surface area of radiating cylinder (m²)
- T_1 is the surface temperature of cylinder (°C)
- T_0 is 273 °C, the constant converting temperature (°C) to absolute temperature (K)
- T_2 is the temperature of media surrounding the cylinder (°C)

Although radiation heat transfer is dependent on a fourth power temperature differential, it can be linearized for relatively small temperature differences and treated as an effective convective heat transfer coefficient as shown in Formula (E.10) and Formula (E.11):

$$h_r = 4 \times \sigma \times \varepsilon \times T_m^3 \quad (\text{E.10})$$

where

- h_r is the linearized radiation heat transfer coefficient (W/m² °C)
- T_m the estimated mean absolute temperature between the surroundings and the radiating surface (K)

$$T_m = T_0 + \frac{T_1 + T_2}{2} \quad (\text{E.11})$$

where the variables are defined as in Formula (E.9).

Example heat loss calculations:

Pipe size: 3,5 in schedule 40 mild steel pipe.

Insulation: Cellular glass, 38 mm thick, with oxidized aluminum weather barrier (oversized to ASTM Standard 4 in nominal pipe insulation size to accommodate a 10 mm outer diameter electric tracer).

- T_p is 65 °C
- T_a is –18 °C
- Wind is 10 m/s
- D_2 is 0,194 m¹
- D_1 is 0,116 m

Emittance of oxidized aluminum weather barrier is 0,11

¹ It is important to note that ASTM Standard thicknesses are nominal and may vary depending on pipe size and nominal insulation thicknesses. Therefore, it is important to know if the insulation thickness being specified is per ASTM C585 or if it is the actual thickness.

Emittance of pipe is 0,9

Emittance of insulation is 0,9

Thermal conductivity of insulation at 24 °C mean is 0,0562 W/m °C

Find: The heat loss per unit length of pipe in both metric and English units, for a horizontal pipe, using:

- a) Formula (E.4) (which results in the highest calculated heat loss)
- b) Formula (E.3)
- c) Formula (E.2)

Example 1 using Formula (E.4):

$$q = \frac{2 \times \pi \times K \times (T_p - T_a)}{\ln(D_2 / D_1)}$$

$$q = \frac{(2) \times (\pi) \times (0,0562) \times (65 - (-18))}{\ln(0,94 / 0,116)}$$

$$q = 56.96 \text{ W/m}$$

Example 2 using Formula (E.3):

$$q = \frac{T_p - T_a}{\frac{\ln(D_2 / D_1)}{2 \times \pi \times K} + \frac{1}{\pi \times D_2 \times h_o}}$$

$$q = \frac{(65 - (-18))}{\frac{\ln(0,194 / 0,116)}{2 \times (\pi) \times (0,0562)} + \frac{1}{(\pi) \times (0,194) \times (52,91)}}$$

$$q = \frac{(83)}{1,457 + 0,031}$$

$$q = 55.78 \text{ W/m}$$

Example 3 using Formula (E.2):

$$q = \frac{T_p - T_a}{\frac{\ln(D_2 / D_1)}{2 \times \pi \times K} + \frac{1}{\pi \times D_2 \times h_{co}} + \frac{1}{\pi \times D_2 \times h_o}}$$

h_{co} is the combination of natural convection and radiation. Using Formula (E.7) and Formula (E.10), and assuming a 6 °C temperature differential between the outer insulation surface and the weather barrier.

$$h_{co} = 1.32 \times \left[\frac{-6 - (-12)}{0.194} \right]^{0.25} + 4(0.9) \times (5.669 \cdot 10^{-8}) \times \left[273 + \frac{-6 + (-12)}{2} \right]^3$$

$$= 3,11 + 3,76$$

$$= 6,87$$

h_o is the combination of forced convection and radiation. Using Formulae (8) and (10), and assuming an 8 °C temperature differential between the weather barrier and the ambient temperature.

$$= h_f + h_r$$

The properties of air evaluated at the film temperature are as follows:

$$k_f = 0,0228 \text{ W/m } ^\circ\text{C}$$

$$\nu_f = 1,07 (10^{-5} \text{ m/s})$$

$$\text{Pr} = 0,72$$

$$h_o = \frac{0.0266 \times (0.0228)}{0.194} \times \left[\frac{11.2 \times (0.194)}{1.07 \times 10^{-5}} \right]^{0.805} (0.72)^{1/3} + 4 \times (0.11) \times (5.669 \cdot 10^{-8}) \times \left[273 + \frac{-12 + (-20)}{2} \right]^3$$

$$= 52,49 + 0,423$$

$$= 52,91$$

From Formula (E.2):

$$q = \frac{(65 - (-18))}{\frac{\ln(0.194 / 0.116)}{2 \times (\pi) \times (0.0562)} + \frac{1}{(\pi) \times (0.194) \times (52.91)}}$$

$$q = \frac{83}{1,457 + 0,239 + 0,031}$$

$$q = 48.06 \text{ W/m}$$

A more in depth analysis of the preceding calculations may be obtained using iterative techniques to determine the thermal conductivity of the thermal insulation and properties of the air film. This would be accomplished by determining what the actual temperature drop across the various thermal resistances would be. However, because these properties are not strong functions of temperature for the small temperature changes that would be found in this example, the total heat loss would be affected minimally.

Annex F (informative)

Vessel heat loss considerations

F.1 General

Unlike pipes, vessel heat losses are affected by heat sinks that are integral to the vessel body and shall be considered during the basic calculation. To determine the total heat loss, the calculations are broken into different regions. The total is derived by summing the regional heat losses as shown in Formula (F.1):

$$Q_{\text{total}} = Q_{\text{ins}} + Q_{\text{slab}} + Q_{\text{supt}} + Q_{\text{manhole}} \quad (\text{F.1})$$

Because of the huge varieties of geometries and fluids, taking a strict theoretical approach to calculation of all possible heat loss rates results in a complex procedure. Most tank heating applications do not require such precision. The following formulae provides a conservative solution for vessel heat loss.

F.2 Insulation heat loss (Q_{ins})

The heat loss through the insulated vessel wall (Q_{ins}) is calculated with Formula (F.2). The worst-case heat loss assumption is made – that the vessel is full. Calculations for partially filled tanks are beyond the scope of this annex. In addition, the fluid film layer on the inside of the tank wall is ignored.

$$Q_{\text{ins}} = \left(\frac{T_p - T_a}{\frac{1}{h_i} + \frac{x}{k} + \frac{1}{h_{\text{co}}} + \frac{1}{h_o}} \right) \times A \quad (\text{F.2})$$

where

- Q_{ins} is the heat loss for region (W)
- A is the tank surface area of insulated region (m^2)
- T_p is the maintain temperature ($^{\circ}\text{C}$)
- T_a is the minimum ambient temperature ($^{\circ}\text{C}$)
- x is the thermal insulation thickness (m)
- k is the thermal insulation conductivity at mean temperature ($\text{W}/\text{m}\cdot^{\circ}\text{C}$)
- h_i is the inside air contact coefficient from the tank to the inside insulation surface ($\text{W}/\text{m}^2\cdot^{\circ}\text{C}$)
- h_{co} is the inside air contact coefficient from the insulation outer surface to the weather barrier ($\text{W}/\text{m}^2\cdot^{\circ}\text{C}$)
- h_o is the outside air film coefficient from the weather barrier or insulation outer surface to the ambient temperature ($\text{W}/\text{m}^2\cdot^{\circ}\text{C}$)

F.3 Slab surface areas (Q_{slab})

If a tank rests directly on a concrete slab, then the heat loss through the bottom of the tank and the concrete slab is calculated in a similar fashion to “wet” areas. For slabs, the term T_{slab} is taken to be the temperature at the slab soil interface, and it replaces the minimum ambient temperature (T_a). Unfortunately, this temperature varies depending on the location under the slab because of the influence of two separate temperatures: the minimum ambient and the deep soil (T_{soil}). The temperature near the edge of the slab generally is lower than the temperature near the middle because the minimum ambient temperature has more of an effect near the outer edges.

When computers are available, the preferred approach to calculating this temperature involves performing a simple numerical nodal temperature analysis for various locations along the slab-soil interface. This approach relies on the fact that at steady state, there is no net heat transfer from any node and a series of simultaneous formulae can be generated. These formulae can then be solved for the temperature of each node using matrix math techniques. Once the temperature of each node is known, Formula (F.3) can be used to calculate the heat loss for that region of the slab:

$$Q_{\text{node}} = \left(\frac{T_p - T_{\text{node}}}{\frac{x_{\text{wall}}}{K_{\text{wall}}} + \frac{x_{\text{slab}}}{k_{\text{slab}}}} \right) \times A_{\text{node}} \quad (\text{F.3})$$

where

- Q_{node} is the heat loss for slab region between nodes (W)
- A_{node} is the surface area of region between nodes (m^2)
- T_p is the maintain temperature ($^{\circ}\text{C}$)
- T_{node} is the calculated nodal temperature ($^{\circ}\text{C}$)
- x_{slab} is the thickness of concrete slab (m)
- k_{slab} is the concrete thermal conductivity at mean temperature ($\text{W}/\text{m}^{\circ}\text{C}$)
- x_{wall} is the tank wall thickness (m)
- k_{wall} is the tank wall conductivity at mean temperature ($\text{W}/\text{m}^{\circ}\text{C}$)

The total heat loss through the slab would then be obtained by summing the nodal heat losses.

There are several analytic alternatives to the computerized nodal temperature analysis approach discussed previously. The analytic approach is usually based on breaking the slab heat loss into two components:

- a) The outer region, which treats the concrete slab as a fin, see Formula (F.4).
- b) The inner region, which uses an formula similar to Formula (F.3).

Typically, the analytic method yields higher heat losses than the nodal method.

F.4 Support heat loss (Q_{supt})

Supports or any other appurtenances that are in contact with the tank wall and extend through the thermal insulation are treated as infinite fins, and their heat loss is calculated using the generalized form of the fin formula as shown in Formula (F.4):

$$Q_{\text{supt}} = \sqrt{h_f \times P \times k_s \times A_c} \times (T_p - T_a) \times \varepsilon \quad (\text{F.4})$$

where

- Q_{supt} is the heat loss for individual support (W)
- A_c is the cross-sectional area of the section that protrudes through the thermal insulation (m²)
- P is the perimeter of A above (m)
- T_p is the maintain temperature (°C)
- T_a is the minimum ambient temperature (°C)
- k_s is the thermal conductivity of support (W/m°C)
- h_f is the convection coefficient from the exposed support surface to the ambient air (W/m² °C)
- ε is the efficiency of the fin. This is a user-definable value. In most cases, it should be set to 1,0.

F.5 Manhole heat loss (Q_{manhole})

If manholes or hand holds are in contact with the fluid, then their heat loss can be determined by using Formula (F.1). If they are not in contact with the fluid, they can safely be ignored.

F.6 Convection coefficient formulae

F.6.1 General

Convection coefficients are difficult to predict accurately because of the wide variety of geometric and fluid conditions encountered with tank heating. Fortunately, for insulated tanks, the effect that many of the coefficients have on overall heat loss is small. To simplify calculation of these coefficients, only three formulae are used. Note that heat transfer from radiation is included in the convection coefficient calculation, thus yielding an effective heat transfer coefficient.

F.6.2 Free convection, nonfluid surface, any orientation (h_i , h_{co} , h_o)

It is assumed that free (or natural) convection takes place whenever the wind speed is less than 0,45 m/s or the surface is enclosed. This is used for h_i and h_{co} for all cases, and for h_o when the wind speed is below 0.45 m/s. Formula (F.5) is based on the vertical wall convection model using constants as presented by Holman [B10]:

$$h_{\text{free}} = \frac{0.1 \times \left(\left(\frac{g \times (T_w - T_\infty) \times L^3}{\nu_{\text{air}}^2 \times T_f} \right) \times \text{Pr} \right)^{1/3} \times k_{\text{air}}}{L} + h_r \quad (\text{F.5})$$

where

- h_{free} is the effective free convection coefficient (W/m²°C)
- k_{air} is the thermal conductivity of air evaluated at the mean film temperature (W/m°C)
- L is the characteristic length (m). Defined as follows:
 - Vertical cylinders: height/2
 - Horizontal cylinders: diameter/2

	Rectangular:	height/2
	Spherical:	diameter/2
g	is the acceleration of gravity (m/s^2)	
T_w	is the wall temperature ($^{\circ}\text{C}$)	
T_{∞}	is the bulk air temperature ($^{\circ}\text{C}$)	
T_f	is the mean film temperature $(T_w - T_{\infty})/2$	
ν_{air}	is the kinematic viscosity of air evaluated at the mean film temperature (m^2/s)	
Pr	is the Prandtl number of air at the mean film temperature (dimensionless)	
h_r	is the radiation component; see Formula (F.7)	

F.6.3 Forced convection, any orientation (h_o)

It is assumed that forced convection takes place whenever the wind speed is greater than 0,45 m/s. This is used for h_o when the wind speed is greater than 0,45 m/s. Formula (F.6) is based on the plate model as presented by Becker [B6]:

$$h_{\text{forced}} = \frac{\Pr^{\frac{1}{3}} \times \left(\left(0.037 \times \frac{V \times L}{\nu_{\text{air}}} \right)^{0.8} \times -871 \right) \times k_{\text{air}}}{L} + h_r \quad (\text{F.6})$$

where

h_{forced}	is the effective forced convection coefficient ($\text{W/m}^2 \text{ } ^{\circ}\text{C}$)
k_{air}	is the thermal conductivity of air evaluated at the mean film temperature ($\text{W/m} \cdot ^{\circ}\text{C}$)
L	is the characteristic length (m). Defined as
	Vertical cylinders: (height + diameter)/2
	Horizontal cylinders: (length + diameter)/2
	Rectangular: (length + width)/2
	Spherical: diameter/2
V	is the wind velocity (m/s)
ν_{air}	is the kinematic viscosity of air evaluated at the mean film temperature (m^2/s)
Pr	is the Prandtl number of air at the mean film temperature (dimensionless)
h_r	is the radiation component; see Formula (E.7)

F.6.4 Radiation component, all coefficients (h_f , h_i , h_{co} , h_o)

Heat transfer from radiation is added to the pure convection value to arrive at an effective convection coefficient. The simplified formula for radiation as presented in Formula (E.10) is used:

$$h_r = 4 \times \sigma \times \varepsilon \times T_m^3 \quad (\text{F.7})$$

where

h_r	is the linearized radiation heat transfer coefficient ($\text{W/m}^2 \text{ } ^{\circ}\text{C}$)
δ	is $5,669 \cdot 10^{-8}$ ($\text{W/m}^2 \cdot \text{K}^4$), the Stefan-Boltzmann constant
ε	is the emissivity of the radiating surface (dimensionless)
T_m	is the estimated mean absolute temperature between the surroundings and the radiating surface (K)

$$T_m = T_0 + \frac{T_1 + T_2}{2} \quad (\text{F.8})$$

where

- T_1 is the surface temperature of the cylinder (°C)
- T_0 is 273 °C, the constant converting temperature (°C) to absolute temperature (K)
- T_2 is the temperature of media surrounding the cylinder (°C)

Annex G (informative)

Heat up and cool down considerations

G.1 Heat up

In certain plant operations, it may be necessary to specify that the heat tracing system be capable of increasing the temperature of a stagnant product to the pumping temperature within a certain time period. Formula (G.1) gives the relationship between heat up time and the heating device input for a pipe:

$$t = H \times \ln \left(\frac{q_c - U \times (T_i - T_a)}{q_c - U \times (T_f - T_a)} \right) + \frac{P_1 \times V_{cl} \times h_f}{q_c - U \times (T_{sc} - T_a)} \quad (G.1)$$

where

U is the heat loss per unit length of pipe per degree of temperature difference; Formula (G.2) is derived from Formula (E.1):

$$U = \frac{1}{\frac{1}{\pi \times D_1 \times h_i} + \frac{\ln(D_2 / D_1)}{2 \times \pi \times K_1} + \frac{\ln(D_3 / D_2)}{2 \times \pi \times K_2} + \frac{1}{\pi \times D_3 \times h_{co}} + \frac{1}{\pi \times D_3 \times h_o}} \quad (G.2)$$

where

H is the thermal time constant, which is the total energy stored in the mass of pipe, fluid, and insulation per degree of temperature divided by the heat loss per unit length per degree temperature differential [see Formula (G.3)]:

$$H = \frac{P_1 \times C_{p1} \times V_{c1} + P_2 \times C_{p2} \times V_{c2} + 0.5 \times P_3 \times C_{p3} \times V_{c3}}{U} \quad (G.3)$$

and

- P_1 is the density of product in the pipe (kg/m³)
- C_{p1} is the specific heat of the product (J/kg °C)
- V_{c1} is the internal volume of the pipe per length (m³/m)
- P_2 is the density of the pipe (kg/m³)
- C_{p2} is the specific heat of the pipe (J/kg °C)
- V_{c2} is the pipe wall volume per length (m³/m)
- P_3 is the density of the insulation (kg/m³)
- C_{p3} is the specific heat of the insulation per length (J/kg °C)
- V_{c3} is the insulation wall volume (m³/m)
- T_i is the initial temperature of the pipe (°C)
- T_f is the final temperature of the fluid and the pipe (°C)
- T_a is the ambient temperature (°C)
- t is the desired heat up time (s)
- U is the heat loss per unit length of pipe per degree of temperature (W/m °C)

H	is the thermal time constant (s)
K_1	is the thermal conductivity of the inner insulation evaluated at its mean temperature (W/m °C)
K_2	is the thermal conductivity of the outer insulation evaluated at its mean temperature (W/m °C)
D_3	is the outside diameter of the outer insulation layer (m)
D_2	is the outside diameter of the inner insulation layer (m)
D_1	is the inside diameter of the insulation layer (m)
h_{co}	is the inside air contact coefficient of the weather barrier (W/m ² °C)
h_o	is the outside air contact coefficient of the weather barrier to the ambient temperature (W/m ² °C)
h_i	is the inside air contact coefficient from the pipe to the inside insulation surface (W/m ² °C)
T_{sc}	is the temperature at which phase change occurs (°C)
h_f	is the latent heat of fusion for the product (J/kg)
q_c	is the heating cable output (W/m)

The preceding relationships also assume that system densities, volumes, thermal conductivities, and heat losses are constant over the temperature range of interest. Note that some products do not undergo a phase change during heat up. Although the model is representative of a straight pipeline, it does not have provisions for equipment such as pumps and valves.

G.2 Cool down

In static (no flow) conditions, with no additional heat from a trace heating device, the length of time to cool down can be calculated from Formula (G.4):

$$t = -H \times \ln \left(\frac{U \times (T_f - T_a)}{U \times (T_i - T_a)} \right) + \frac{P_1 \times V_{cl} \times h_f}{U \times (T_{sc} - T_a)} \quad (G.4)$$

where

U is the heat loss per unit length of the pipe per degree of temperature difference and can be calculated from Formula (G.5):

$$U = \frac{1}{\frac{1}{\pi \times D_1 \times h_i} + \frac{\ln(D_2/D_1)}{2 \times \pi \times K_1} + \frac{\ln(D_3/D_2)}{2 \times \pi \times K_2} + \frac{1}{\pi \times D_3 \times h_{co}} + \frac{1}{\pi \times D_3 \times h_o}} \quad (G.5)$$

H is the thermal time constant; the total energy stored in the mass of the pipe, fluid, and insulation per degree of temperature divided by the heat loss per unit length per degree temperature differential and can be calculated by Formula (G.6):

$$H = \frac{P_1 \times C_{p1} \times V_{cl} + P_2 \times C_{p2} \times V_{c2} + 0.5 \times P_3 \times C_{p3} \times V_{c3}}{U} \quad (G.6)$$

where

P_1	is the density of the product in the pipe (kg/m ³)
C_{p1}	is the specific heat of the product (J/kg °C)

V_{c1}	is the internal volume of the pipe per length (m^3/m)
P_2	is the density of the pipe (kg/m^3)
C_{p2}	is the specific heat of the pipe ($\text{J}/\text{kg } ^\circ\text{C}$)
V_{c2}	is the pipe wall volume per length (m^3/m)
P_3	is the density of the insulation (kg/m^3)
C_{p3}	is the specific heat of the insulation ($\text{J}/\text{kg } ^\circ\text{C}$)
V_{c3}	is the insulation wall volume per length (m^3/m)
T_i	is the initial temperature of the pipe ($^\circ\text{C}$)
T_f	is the final temperature of the fluid and the pipe ($^\circ\text{C}$)
T_a	is the ambient temperature ($^\circ\text{C}$)
t	is the desired heat up time (s)
U	is the heat loss per unit length of the pipe per degree of temperature ($\text{W}/\text{m } ^\circ\text{C}$)
H	is the thermal time constant (s)
K_1	is the thermal conductivity of the inner insulation evaluated at its mean temperature ($\text{W}/\text{m } ^\circ\text{C}$)
K_2	is the thermal conductivity of the outer insulation evaluated at its mean temperature ($\text{W}/\text{m } ^\circ\text{C}$)
D_3	is the outside diameter of the outer insulation layer (m)
D_2	is the outside diameter of the inner insulation layer (m)
D_1	is the inside diameter of the insulation layer (m)
h_{co}	is the inside air contact coefficient of the weather barrier ($\text{W}/\text{m}^2 \text{ } ^\circ\text{C}$)
h_o	is the outside air contact coefficient of the weather barrier to the ambient temperature ($\text{W}/\text{m}^2 \text{ } ^\circ\text{C}$)
h_i	is the inside air contact coefficient from the pipe to the inside insulation surface ($\text{W}/\text{m}^2 \text{ } ^\circ\text{C}$)
T_{SC}	is the temperature at which phase change occurs ($^\circ\text{C}$)
h_f	is the latent heat of fusion for the product (J/kg)

Annex H

(informative)

Method to determine equivalent thicknesses of insulating cements

Given: A pipe insulated with 25,4 mm (actual thickness) mineral wool with a thermal conductivity of 0,0447 W/ m°C at 37,8 °C mean temperature.

Find: The proper thickness of the insulating cement that has been reported by the manufacturer to have a thermal conductivity of 0,0706 W/m·°C at 37,8 °C mean temperature.

Solution, as shown in Formula (H.1):

$$\frac{K_{\text{ins}}}{K_{\text{cement}}} = \frac{t_{\text{ins}}}{t_{\text{cement}}} \quad (\text{H.1})$$

$$\frac{0.0447}{0.0706} = \frac{25.4}{t_{\text{cement}}}$$

$$t_{\text{cement}} = 40,1 \text{ mm}$$

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CSA C22.1, *Canadian Electrical Code, Part 1*

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